

(45) **Date of Patent:** Sep. 9, 2025

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US 2024/0250181 A1 Jul. 25, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/500,020, filed on Oct. 13, 2021, now Pat. No. 11,929,437, which is a
(Continued)

(30) **Foreign Application Priority Data**

Apr. 13, 2012 (JP) 2012-091539

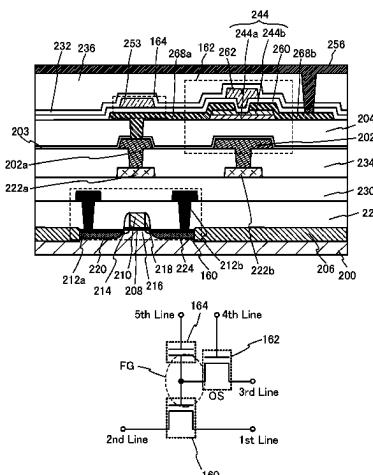
(51) **Int. Cl.**
H10D 30/67 (2025.01)
H10D 30/60 (2025.01)

(52) **U.S. Cl.**
CPC *H10D 30/6755* (2025.01); *H10D 30/611*
(2025.01)

(57) **ABSTRACT**

A transistor includes oxide semiconductor stacked layers between a first gate electrode layer and a second gate electrode layer through an insulating layer interposed between the first gate electrode layer and the oxide semiconductor stacked layers and an insulating layer interposed between the second gate electrode layer and the oxide semiconductor stacked layers. The thickness of a channel formation region is smaller than the other regions in the oxide semiconductor stacked layers. Further in this transistor, one of the gate electrode layers is provided as what is called a back gate for controlling the threshold voltage.

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Controlling the potential applied to the back gate enables control of the threshold voltage of the transistor, which makes it easy to maintain the normally-off characteristics of the transistor.

10 Claims, 13 Drawing Sheets

Related U.S. Application Data

continuation of application No. 17/065,635, filed on Oct. 8, 2020, now Pat. No. 11,355,645, which is a continuation of application No. 16/707,432, filed on Dec. 9, 2019, now Pat. No. 10,872,981, which is a continuation of application No. 16/180,210, filed on Nov. 5, 2018, now Pat. No. 10,559,699, which is a continuation of application No. 15/262,547, filed on Sep. 12, 2016, now Pat. No. 10,158,026, which is a continuation of application No. 14/594,991, filed on Jan. 12, 2015, now Pat. No. 9,472,679, which is a continuation of application No. 13/860,792, filed on Apr. 11, 2013, now Pat. No. 8,946,702.

(58) Field of Classification Search

USPC 257/43
See application file for complete search history.

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FIG. 1A

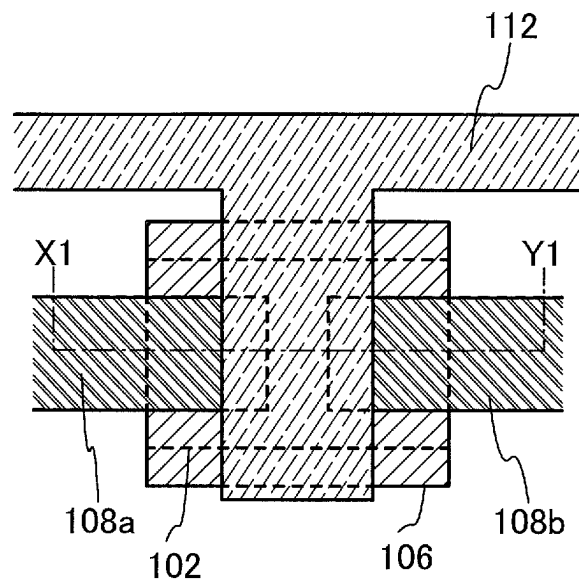
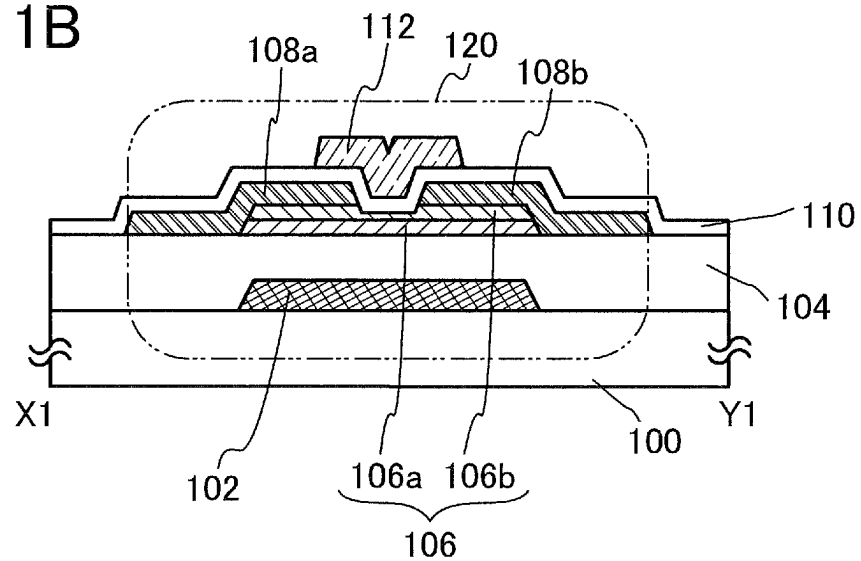


FIG. 1B



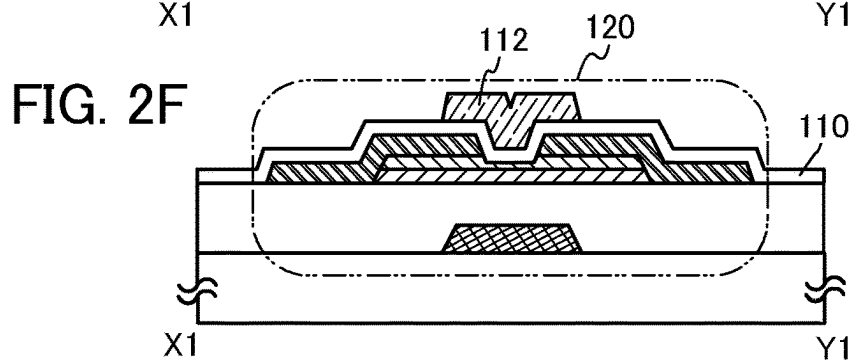
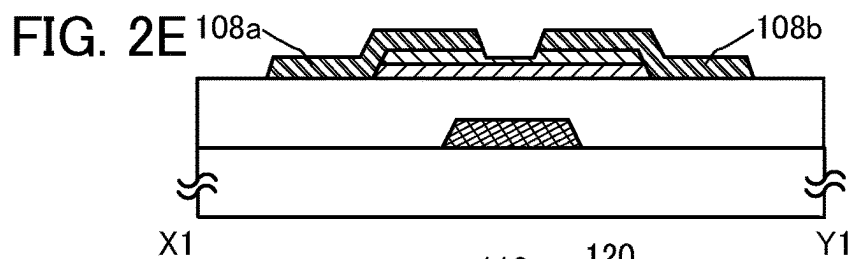
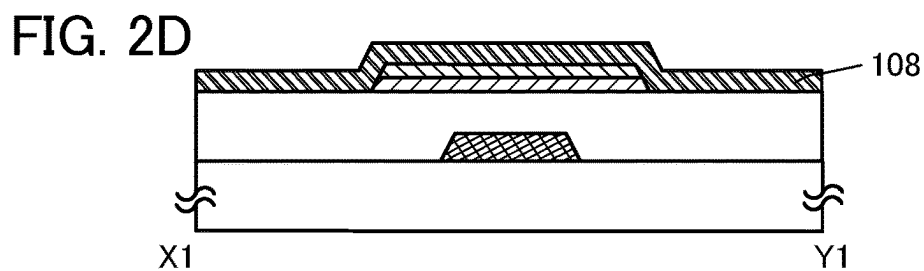
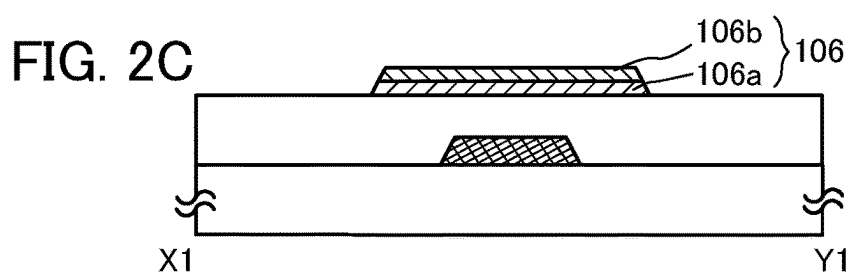
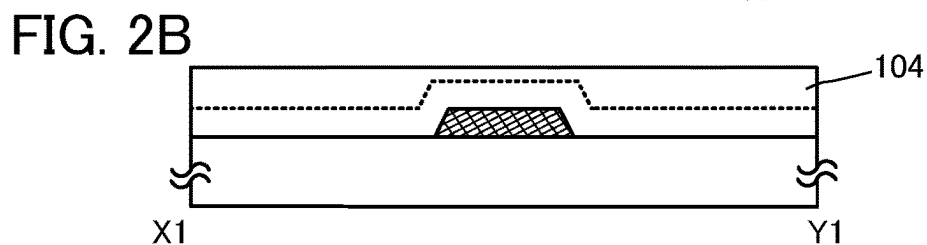
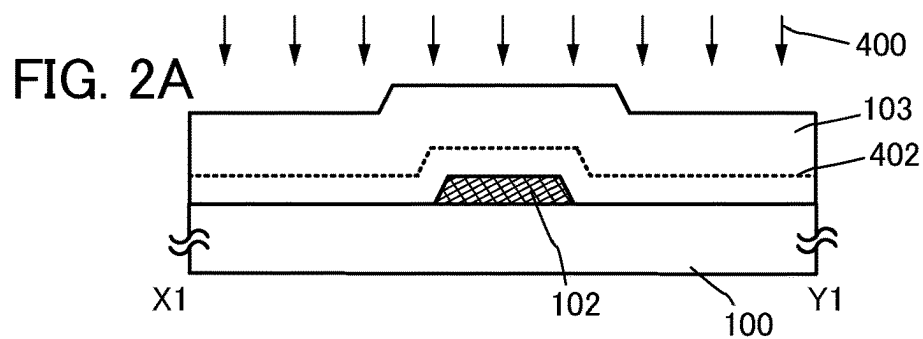


FIG. 3A

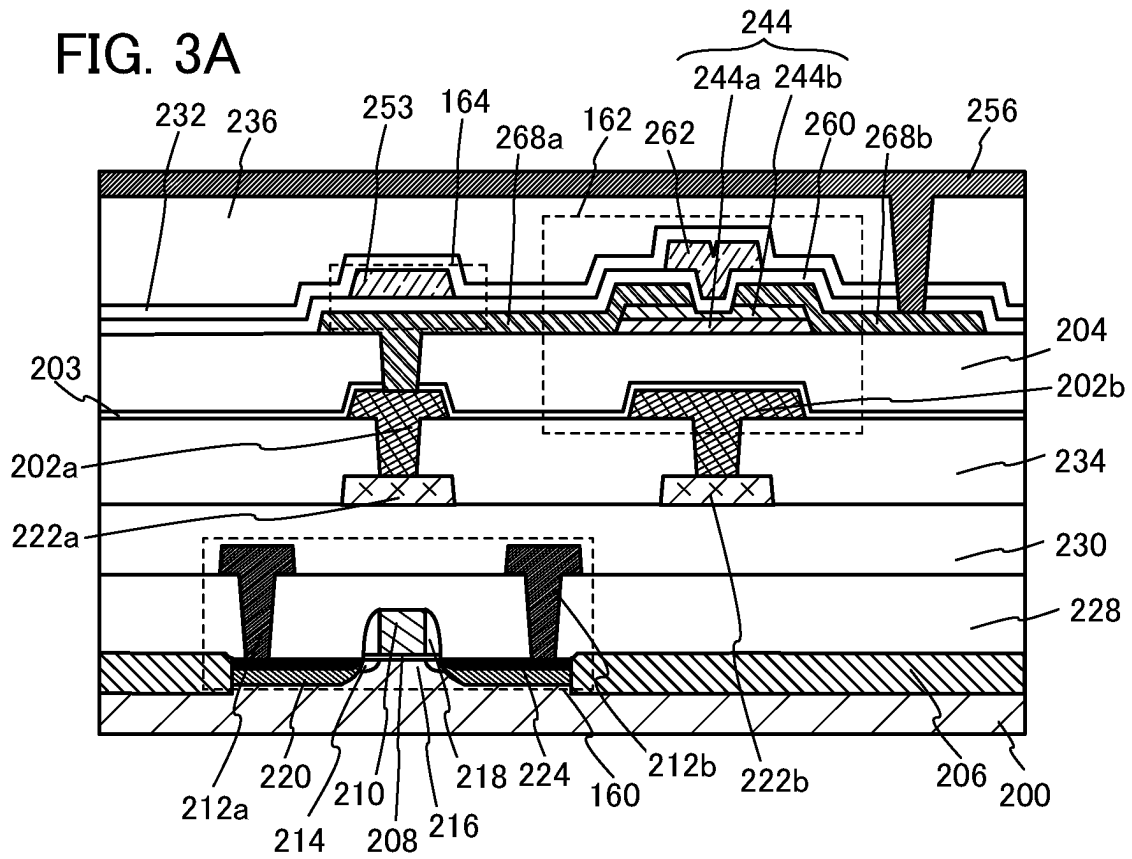


FIG. 3B

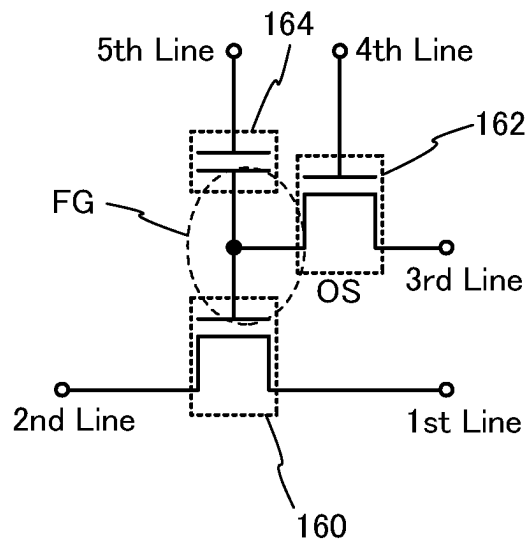


FIG. 4A

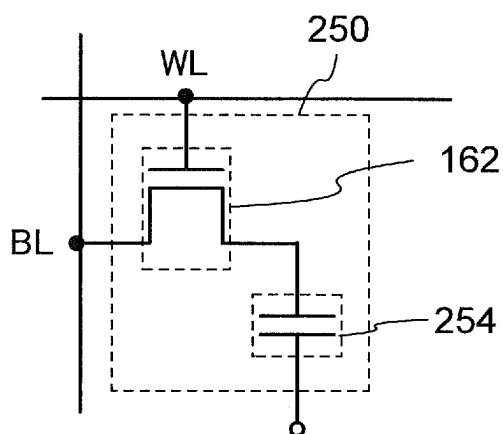


FIG. 4B

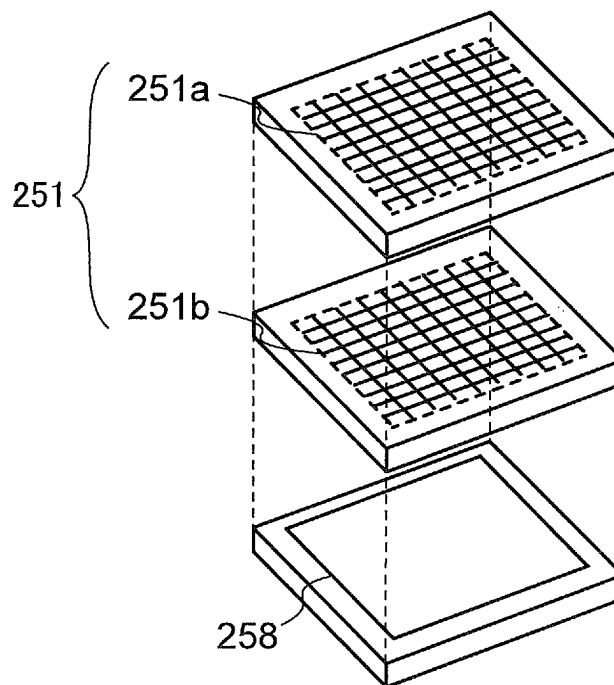


FIG. 5A

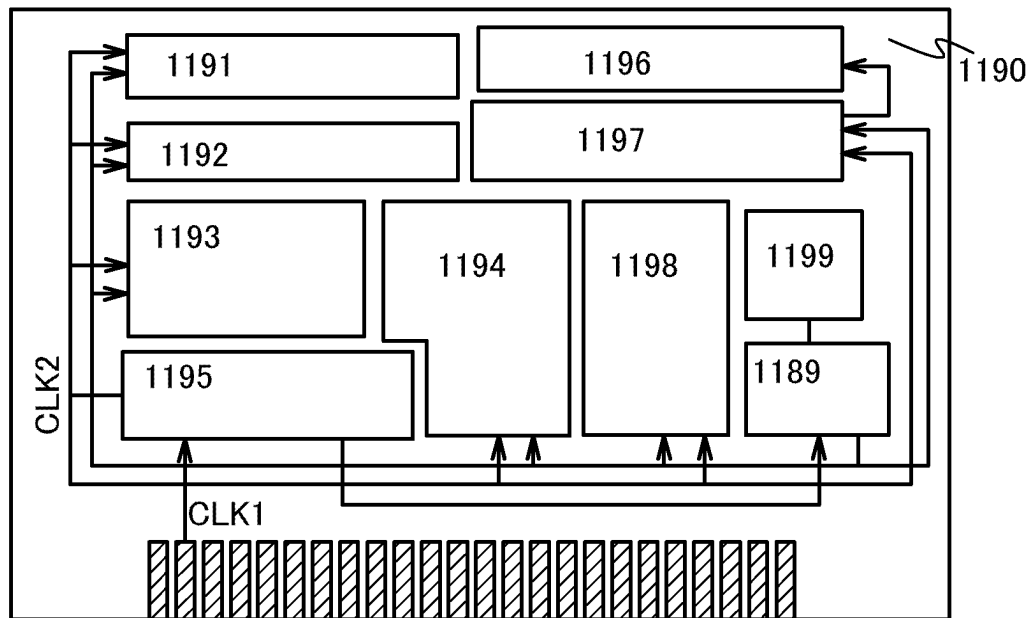


FIG. 5B

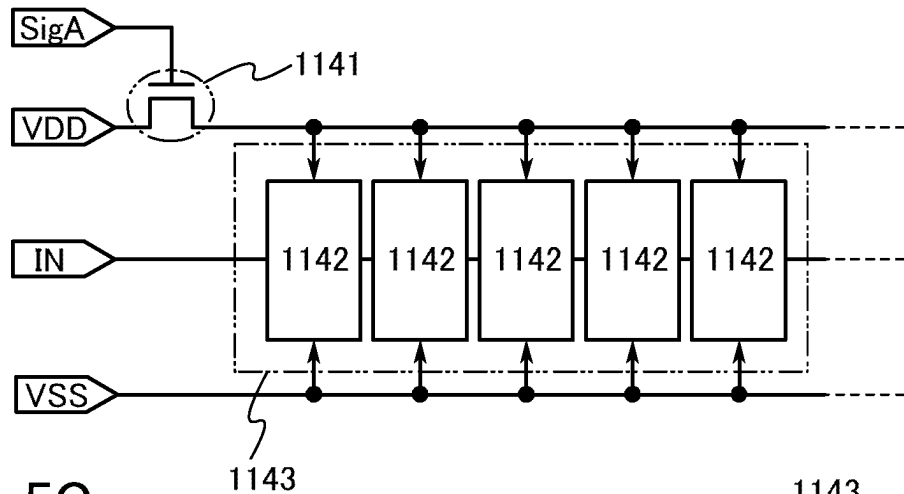


FIG. 5C

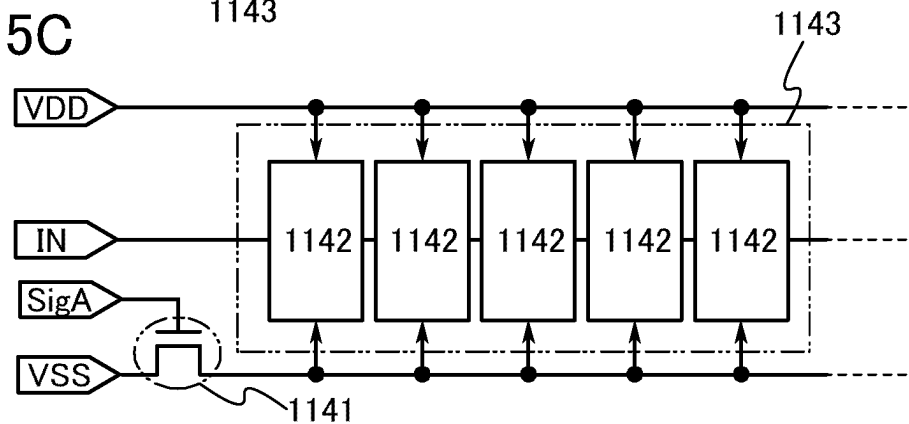


FIG. 6A

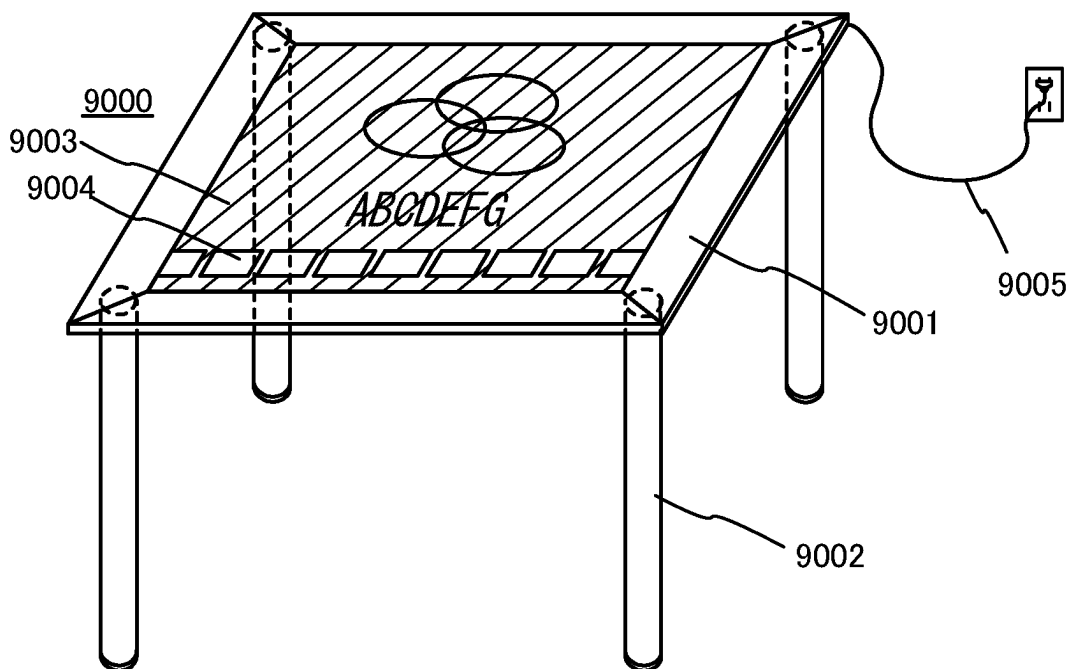


FIG. 6B

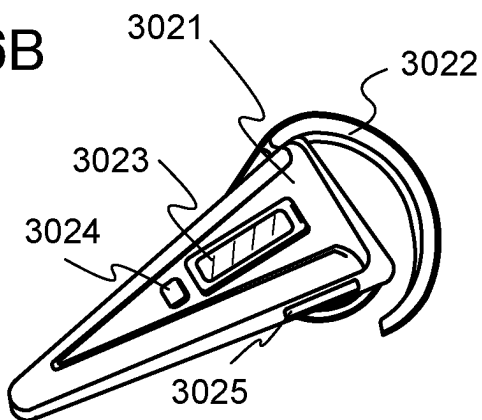


FIG. 6C

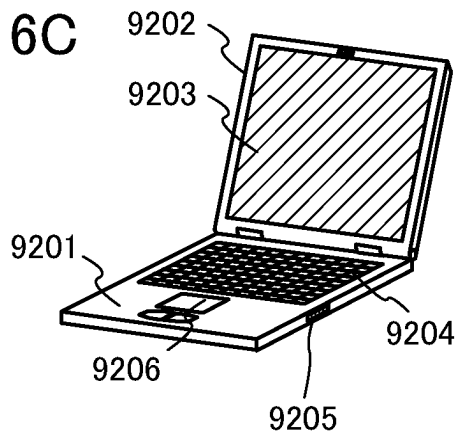


FIG. 7A

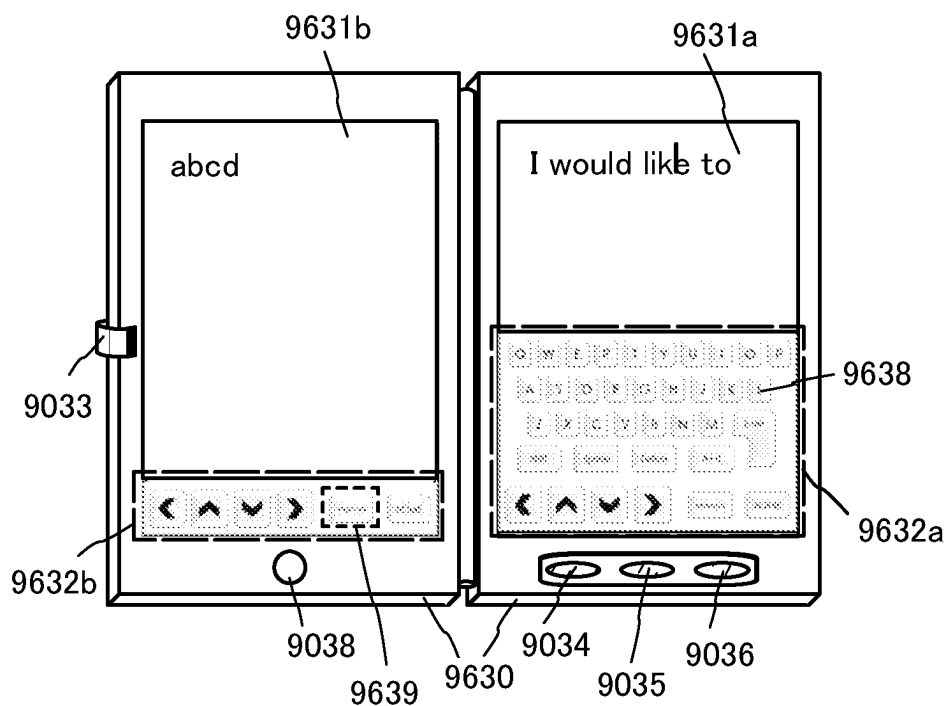


FIG. 7B

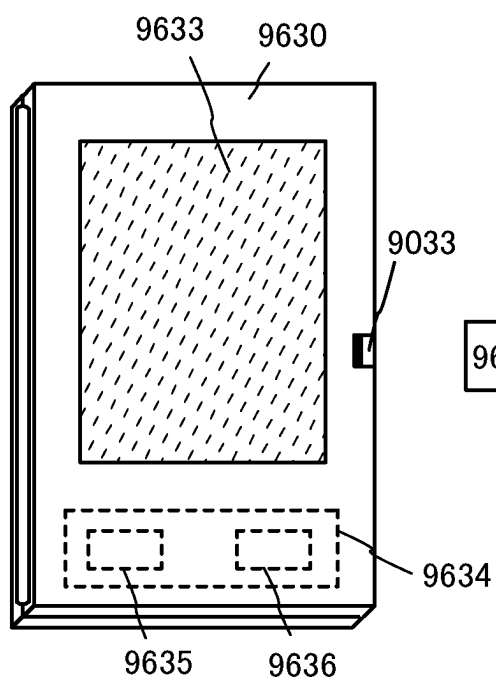


FIG. 7C

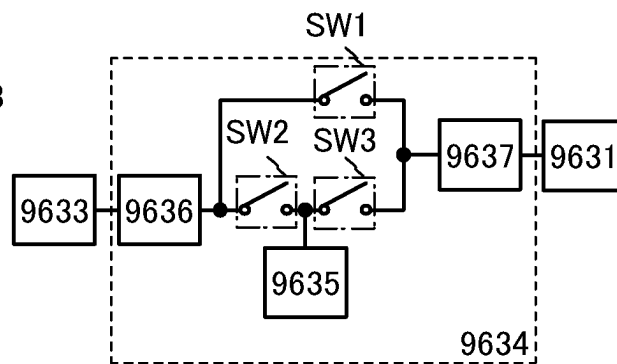


FIG. 8A

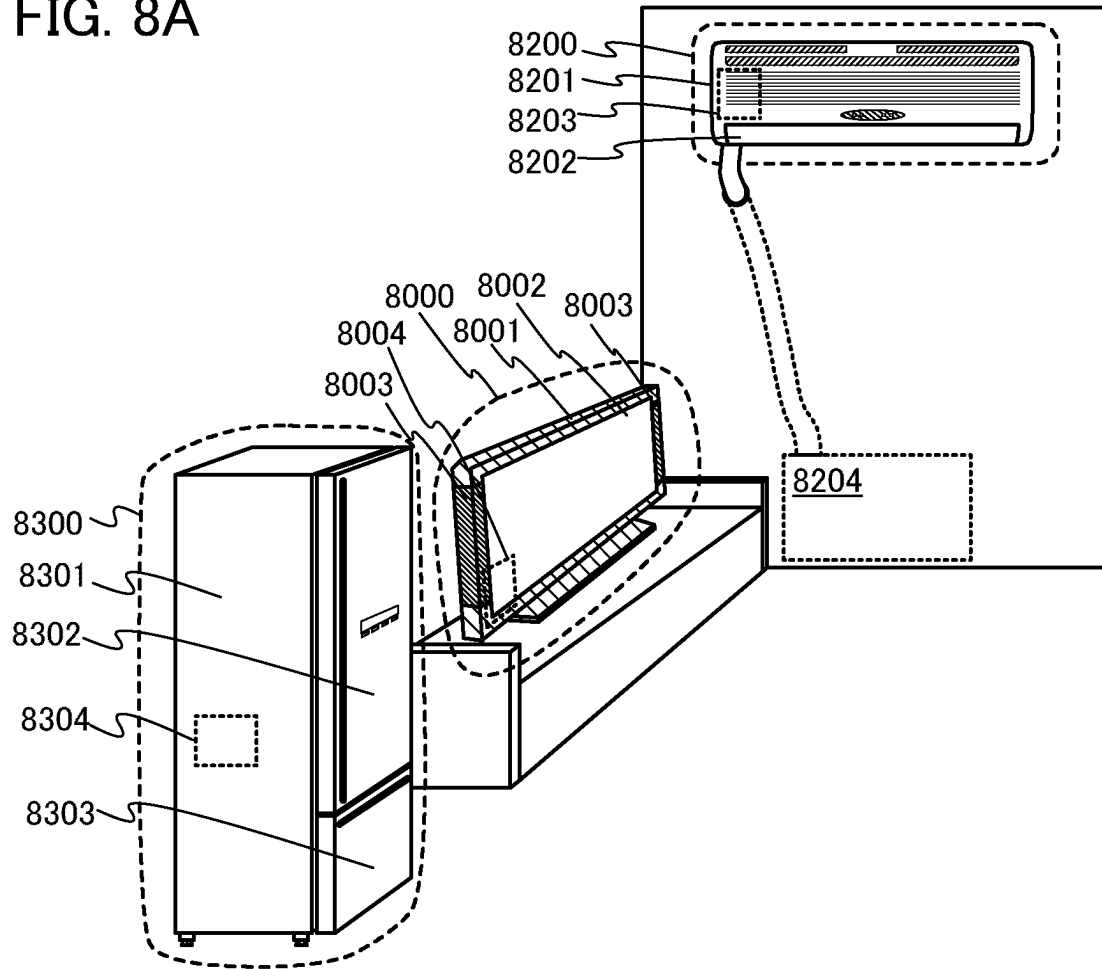


FIG. 8B

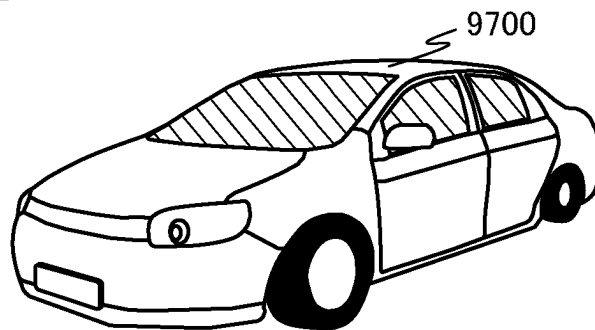


FIG. 8C

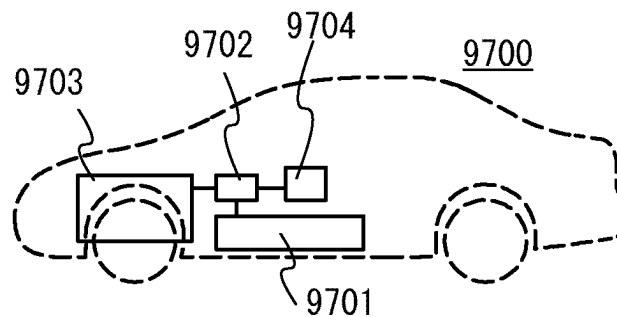


FIG. 9

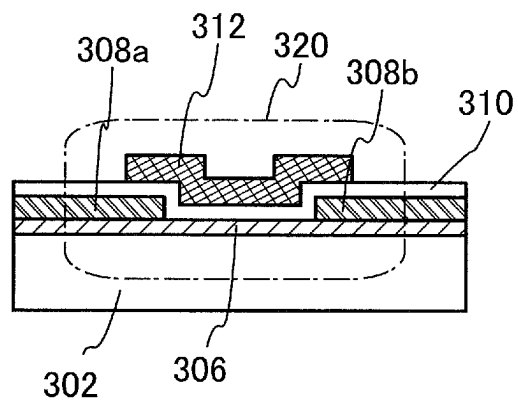


FIG. 10A

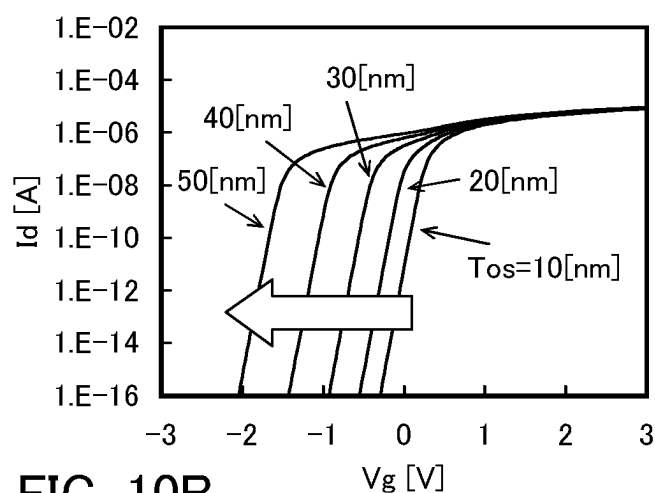


FIG. 10B

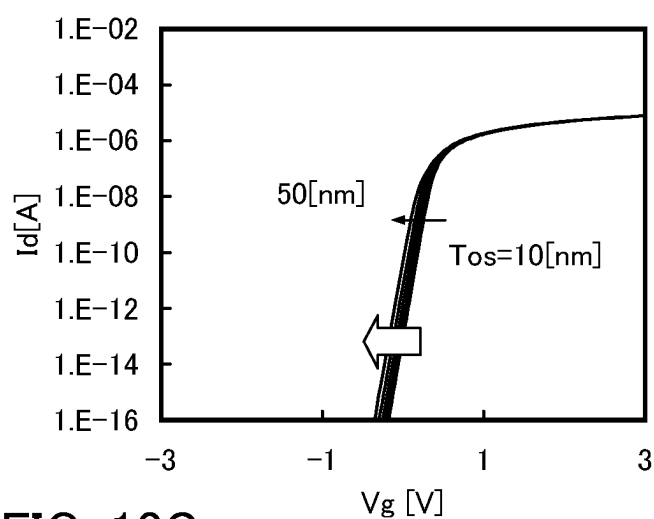


FIG. 10C

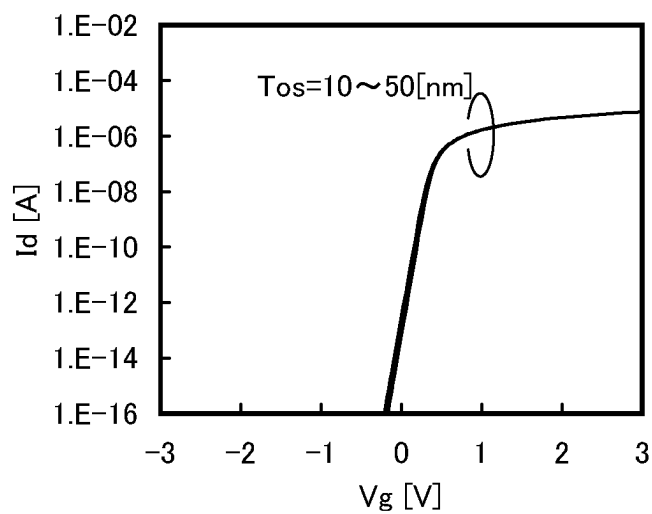


FIG. 11A

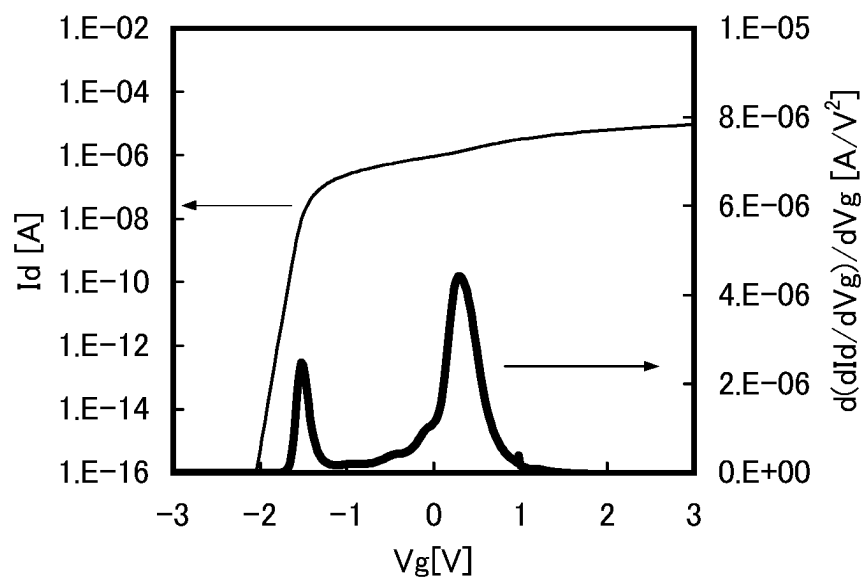


FIG. 11B

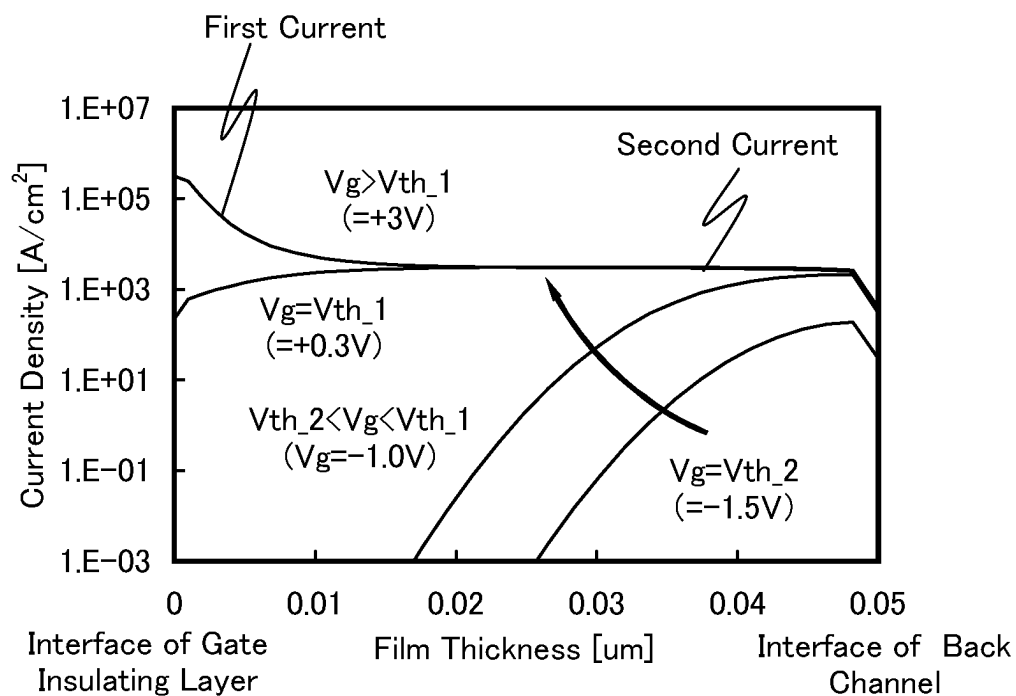
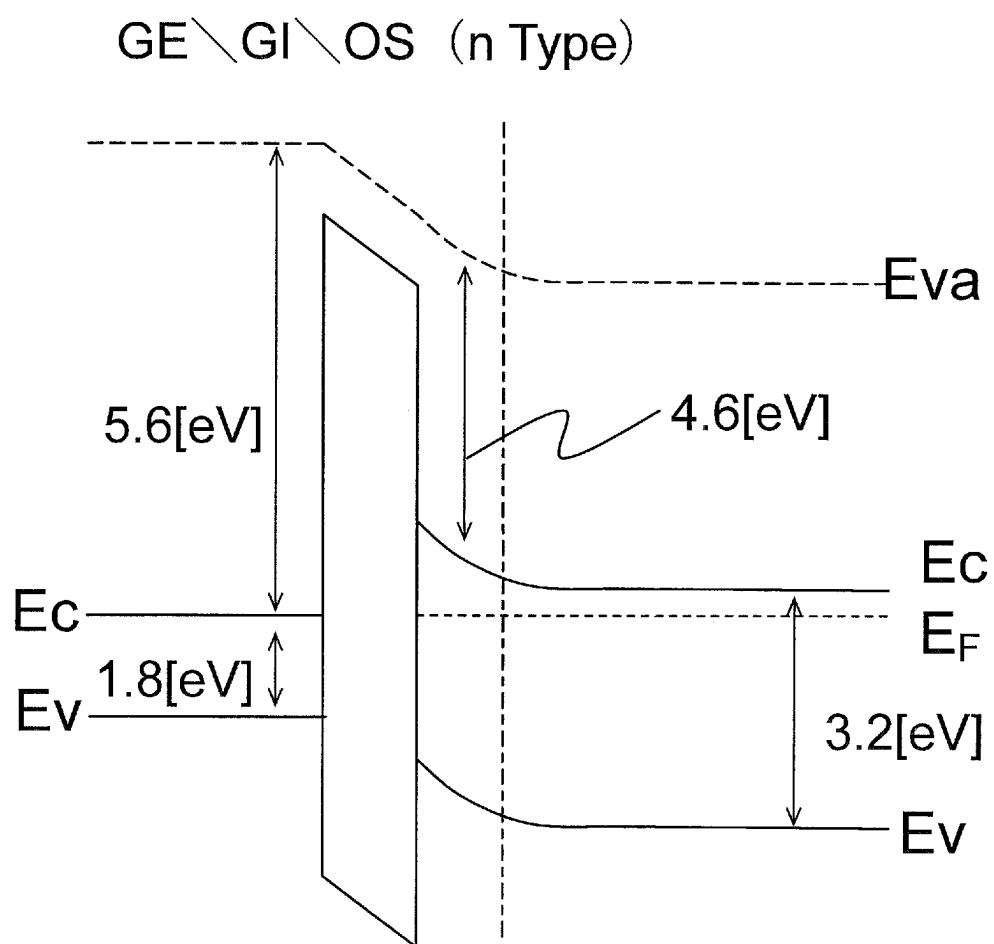
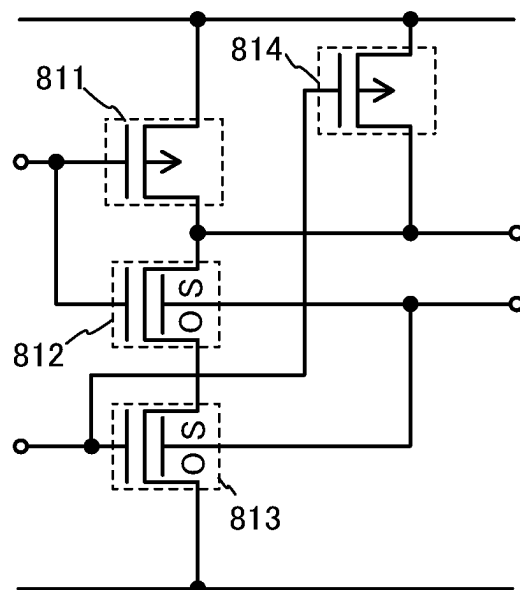


FIG. 12





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SEMICONDUCTOR DEVICE COMPRISING CONDUCTIVE LAYERS FUNCTIONING AS FIRST AND SECOND GATE ELECTRODES OF A TRANSISTOR

TECHNICAL FIELD

The invention disclosed in this specification relates to a semiconductor device and a method for manufacturing the semiconductor device.

In this specification and the like, a semiconductor device refers to all types of devices that can function by utilizing semiconductor characteristics; an electro-optical device, a light-emitting display device, a semiconductor circuit, and an electronic device are all semiconductor devices.

BACKGROUND ART

A technique for forming a transistor by using a semiconductor thin film formed over a substrate having an insulating surface has attracted attention. Such a transistor is applied to a wide range of semiconductor electronic devices such as an integrated circuit (IC) and an image display device (also simply referred to as display device). A silicon-based semiconductor material is widely known as a material for a semiconductor thin film applicable to the transistor. As another material, an oxide semiconductor has been attracting attention.

For example, a technique is disclosed by which a transistor is manufactured using zinc oxide or an In—Ga—Zn-based oxide as an oxide semiconductor (see Patent Documents 1 and 2).

REFERENCE

Patent Document

[Patent Document 1] Japanese Published Patent Application No. 2007-123861

[Patent Document 2] Japanese Published Patent Application No. 2007-096055

DISCLOSURE OF INVENTION

In a transistor used in a semiconductor device, it is preferable that a channel be formed at a positive threshold voltage (V_{th}) which is as close to 0 V as possible. A transistor having a negative threshold voltage tends to be what is called a normally-on transistor that passes a current between a source and a drain even at a gate voltage of 0 V, and a circuit including such a transistor is difficult to control. For this reason, the transistor having the negative threshold voltage is not suited for the use in an integrated circuit of a semiconductor device.

In view of the above, an object of one embodiment of the present invention is to provide a structure of an n-channel transistor including an oxide semiconductor in its channel formation region which has a positive threshold voltage, that is, which enables a normally-off switching element, and to provide a method for forming the structure.

Further, it is important to make transistor characteristics close to the normally-off characteristics even when the transistor cannot become a normally-off transistor due to the material or the manufacturing condition. Thus, another object of one embodiment of the present invention is to provide a structure of a transistor which enables a threshold voltage close to zero even when the threshold voltage is

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negative, that is, even when the transistor is a normally-on transistor, and to provide a method for forming the structure.

Note that one embodiment of the present invention achieves at least one of the above objects.

In one embodiment of the present invention, a transistor includes oxide semiconductor stacked layers between a first gate electrode layer and a second gate electrode layer through an insulating layer interposed between the first gate electrode layer and the oxide semiconductor stacked layers and an insulating layer interposed between the second gate electrode layer and the oxide semiconductor stacked layers, and the thickness of a channel formation region is smaller than the other regions in the oxide semiconductor stacked layers. Further in this transistor, one of the gate electrode layers is provided as what is called a back gate for controlling the threshold voltage. Controlling the potential applied to the back gate enables control of the threshold voltage of the transistor, which makes it easy to maintain the normally-off characteristics of the transistor. More specifically, the following structures can be employed, for example.

One embodiment of the present invention is a semiconductor device which includes a first gate electrode layer over an insulating surface, a first insulating layer over the first gate electrode layer, oxide semiconductor stacked layers including a first oxide semiconductor layer and a second oxide semiconductor layer and overlapping with the first gate electrode layer with the first insulating layer interposed between the first gate electrode layer and the oxide semiconductor stacked layers, a source electrode layer and a drain electrode layer over and in contact with the second oxide semiconductor layer, a second insulating layer over and in contact with the source electrode layer, the drain electrode layer, and part of the oxide semiconductor stacked layers, and a second gate electrode layer overlapping with the oxide semiconductor stacked layers with the second insulating layer interposed therebetween. In the oxide semiconductor stacked layers, a region in contact with the second insulating layer has a smaller thickness than a region in contact with the source electrode layer and a region in contact with the drain electrode layer.

Another embodiment of the present invention is a semiconductor device which includes a first gate electrode layer over an insulating surface, a first insulating layer over the first gate electrode layer, oxide semiconductor stacked layers including a first oxide semiconductor layer and a second oxide semiconductor layer and overlapping with the first gate electrode layer with the first insulating layer interposed between the first gate electrode layer and the oxide semiconductor stacked layers, a source electrode layer and a drain electrode layer over and in contact with the second oxide semiconductor layer, a second insulating layer over and in contact with the source electrode layer, the drain electrode layer, and part of the oxide semiconductor stacked layers, and a second gate electrode layer overlapping with the oxide semiconductor stacked layers with the second insulating layer interposed therebetween. In the semiconductor device, the first oxide semiconductor layer and the second oxide semiconductor layer have the same constituent elements and different compositions of the constituent elements. Further, in the oxide semiconductor stacked layers, a region in contact with the second insulating layer has a smaller thickness than a region in contact with the source electrode layer and a region in contact with the drain electrode layer.

In the above-described semiconductor devices, it is preferable that the first oxide semiconductor layer include at

least indium and gallium, and that an indium content be higher than a gallium content in the first oxide semiconductor layer.

Further in the above-described semiconductor devices, it is preferable that the second oxide semiconductor layer include at least indium and gallium, and that an indium content be lower than or equal to a gallium content in the second oxide semiconductor layer.

Further in the above-described semiconductor devices, it is preferable that at least one of the first gate electrode layer and the second gate electrode layer be a conductive layer having a work function of 5 electron volts or more. For example, at least one of the first gate electrode layer and the second gate electrode layer is preferably an In—Ga—Zn—O film including nitrogen.

The effect of the above-described structures of embodiments of the invention disclosed herein can be explained as follows. Note that the following description is merely one consideration.

A transistor using an oxide semiconductor can be regarded as an accumulation-mode n-channel MOSFET using electrons which are its majority carriers. In an n-channel inversion-mode MOSFET using silicon, an inversion layer is formed in the vicinity of a surface of an active layer (silicon here) by application of a gate voltage, thereby forming a channel. Meanwhile in the accumulation-mode MOSFET, a channel through which a current flows is formed by accumulation of electrons, which are majority carriers, in the vicinity of a surface of an active layer (an oxide semiconductor layer here) in an on state. Further, in an off state, the entire active layer is completely depleted by being applied with a negative gate voltage.

As the on-state current in the accumulation-mode MOSFET, there exist a first current that flows in the vicinity of the surface (accumulation) and a second current that flows through the entire region in the film thickness direction in the active layer. This is largely different from the inversion-mode MOSFET. Here, assuming that the threshold voltage of the first current is V_{th1} and the threshold voltage of the second current is V_{th2} , when the gate voltage V_g is lower than the threshold voltage of the second current ($V_g < V_{th2}$), the entire region in the film thickness direction in the active layer is depleted (completely depleted) and the transistor is in an off state. When the gate voltage V_g is increased to be higher than the threshold voltage V_{th2} of the second current and lower than the threshold voltage V_{th1} of the first current ($V_{th2} < V_g < V_{th1}$), the width of the depletion layer decreases (partly depleted state), and the second current flows on the back channel side; thus, the transistor is in an on state. When the gate voltage V_g is further increased to be a voltage exceeding the threshold voltage of the first current ($V_{th1} < V_g$), the depletion layer disappears and carriers (electrons) are accumulated in the vicinity of the surface of the active layer; thus, the first current flows.

The threshold voltage V_{th2} of the second current and the threshold voltage V_{th1} of the first current in the accumulation-mode n-channel MOSFET are expressed by Equation 1 and Equation 2 using gradual channel approximation.

$$V_{th2} = V_{FB} - \left(1 + 2 \frac{C_S}{C_{ox}}\right) \frac{eN_d t_S^2}{2\epsilon_S} \quad [\text{Equation 1}]$$

$$= V_{FB} - \left(\frac{C_{ox}}{C_S} + 2\right) \frac{eN_d t_S^2}{2C_{ox}}$$

$$V_{th1} = V_{FB} \quad [\text{Equation 2}]$$

In Equations 1 and 2, V_{FB} is a flat band voltage, C_S is a capacitance of the active layer, C_{ox} is a capacitance of a gate insulating layer, N_d is a donor density, and t_S is a thickness of the active layer.

According to Equation 1, increase in the donor density (N_d) and the thickness (t_S) of the active layer moves (shifts) the threshold voltage V_{th2} of the second current in the negative direction. Further, according to Equation 1 and Equation 2, increase in the flat band voltage (V_{FB}), that is, increase in the work function of a gate electrode layer is important in shifting the threshold voltages (V_{th1} and V_{th2}) of the accumulation-mode n-channel MOSFET in the positive direction.

Next, dependence of electric characteristics of a transistor that uses an oxide semiconductor on the thickness (T_{OS}) of an oxide semiconductor layer and the donor density (N_d) was calculated with a device simulator.

The transistor structure assumed in the calculation is shown in FIG. 9. Further, calculation conditions are shown in Table 1.

This calculation used a transistor 320 illustrated in FIG. 9, which includes an oxide semiconductor layer 306 formed over an insulating layer 302 with a thickness of 300 nm, a source electrode layer 308a and a drain electrode layer 308b over the oxide semiconductor layer 306, a gate insulating layer 310 that covers the source electrode layer 308a and the drain electrode layer 308b and is partly in contact with the oxide semiconductor layer 306, and a gate electrode layer 112 that overlaps with the oxide semiconductor layer 306 with the gate insulating layer 310 sandwiched therebetween.

TABLE 1

Channel length (L)/Channel width (W)	1/1 [μm]
Thickness (T_{ox}) of gate insulating layer/Dielectric constant (ϵ)	10 [nm]/22
Thickness (T_{OS}) of oxide semiconductor layer	10, 20, 30, 40, 50 [nm]
Thickness of source electrode layer or drain electrode layer/Work function	50 [nm]/4.6 eV
Work function of gate electrode layer	5.0 [eV]
Parameters of oxide semiconductor	Band gap (E_g) 3.15 [eV] Electron affinity (χ) 4.6 [eV] Dielectric constant (ϵ) 15
Electron mobility (μ_n)	20 [cm^2/Vs]
Hole mobility (μ_p)	0.1 [cm^2/Vs]
Nc	5E18 [cm^{-3}]
Nv	5E18 [cm^{-3}]
Donor density (N_d)	1E16, 1E17, 1E18 [cm^{-3}]

* Defect level and tunneling current are not assumed.

* No fixed charge at the interface on the gate insulating layer side.

The $I_d V_g$ characteristics ($V_d=0.1$ V) obtained by this calculation are shown in FIGS. 10A to 10C. FIGS. 10A to 10C show transistor characteristics based on the assumption that the density (N_a) of donors contained in the oxide semiconductor layer is $1 \times 10^{18} \text{ cm}^{-3}$ (FIG. 10A), $1 \times 10^{17} \text{ cm}^{-3}$ (FIG. 10B), and $1 \times 10^{16} \text{ cm}^{-3}$ (FIG. 10C).

In the case where the donor density (N_d) is high as shown in FIG. 10A, electric characteristics are favorable with a small thickness of the oxide semiconductor layer (e.g., 10 nm); however, normally-on characteristics are observed with larger thicknesses of the oxide semiconductor layer.

In the case where the donor density (N_d) is reduced to $1 \times 10^{17} \text{ cm}^{-3}$ as shown in FIG. 10B, the shift of the characteristics in the negative direction in accordance with the increase in the thickness of the oxide semiconductor layer is smaller. Moreover, the on-state current (I_{on}) is almost uniform without dependence on the thickness of the oxide semiconductor layer. Further, in the case where the donor

density (Na) is reduced to $1 \times 10^{16} \text{ cm}^{-3}$ as shown in FIG. 10C, almost no shift of characteristics in the negative direction in accordance with the increase in the thickness of the oxide semiconductor layer is seen.

From the above-described calculation results, the decrease in the thickness of the oxide semiconductor layer and the decrease in the density of donors contained in the oxide semiconductor layer are the keys to achieving a normally-off transistor.

A transistor described as one embodiment of the present invention includes oxide semiconductor stacked layers in which a channel formation region is thinner than the other region (e.g., a region in contact with the source electrode layer or the drain electrode layer). This can suppress the shift of the threshold voltage of the transistor in the negative direction.

Next, causes of the normally-on characteristics of the transistor are considered. In this consideration, the transistor with a thickness of the oxide semiconductor layer of 50 nm and a donor density of $1 \times 10^{18} \text{ cm}^{-3}$, which has characteristics shown in FIG. 10A, is used.

As described above, in the accumulation-mode MOSFET, there exist the first current that flows in the vicinity of the surface of the active layer (accumulation) and the second current that flows through the entire region in the film thickness direction in the active layer as the on-state current. It is known that the first current and the second current can be distinguished from each other by second-order differentiation of the $I_d V_g$ characteristics. The graph in FIG. 11A shows $I_d V_g$ characteristics (solid line) of the transistor with a thickness of the oxide semiconductor layer of 50 nm and a donor density of $1 \times 10^{18} \text{ cm}^{-3}$, and values obtained by the second-order differentiation (thick line).

In FIG. 11A, the line obtained by the second-order differentiation of the $I_d V_g$ characteristics has two peaks. This indicates that when the gate voltage (V_g) reaches the first peak ($V_g = -1.52 \text{ V}$) by sweeping the gate voltage (V_g) in the positive direction from -3 V , the second current starts flowing and when it reaches the second peak ($V_g = 0.30 \text{ V}$), the first current starts flowing. These numerical results are almost consistent with the calculation results ($V_{th,2} = -1.56 \text{ V}$, $V_{th,1} = 0.36 \text{ V}$) obtained by substituting the parameters in Table 1 into Equation 1 and Equation 2 which use gradual channel approximation.

Thus, it is effective to suppress the second current that flows at a low gate voltage rather than the first current, in order to achieve a normally-off transistor.

FIG. 11B shows current density distributions in a film thickness direction at several gate voltages. When a gate voltage V_g of -3 V is applied, the transistor is in an off state and in a completely depleted state without electrons in the channel region. At the gate voltage V_g higher than $V_{th,2}$, the channel region is in a partly depleted state; at this time, the second current starts flowing on the back channel side. When $V_{th,2} < V_g < V_{th,1}$, the second current is dominant in the on-state current. When the gate voltage V_g is higher than $V_{th,1}$, the second current is not increased and the current density in the vicinity of the interface of the gate insulating layer is increased. At this time, the current density of the second current is about two orders of magnitude smaller than that of the first current. In other words, the first current is dominant when the transistor is in an on state.

The transistor described as one embodiment of the present invention includes a first gate electrode layer and a second gate electrode layer between which the oxide semiconductor layer including the channel formation region is sandwiched. A bias voltage is applied to one of the gate electrode layers

to suppress generation of the second current on the back channel side. In this way, the threshold voltage of the transistor can be moved in the positive direction.

Further, a conductive layer having a large work function (e.g., 5 eV or more) can be used as the gate electrode layers, whereby the threshold voltage can be moved in the positive direction. As the conductive layer having a large work function, an In—Ga—Zn—O film including nitrogen at least at a concentration higher than that of the oxide semiconductor layer can be used, for example.

Note that in the case where the gate insulating layer (the insulating layer provided between the gate electrode layer and the oxide semiconductor layer) contains positive ions such as sodium ions, the positive ions move to the interface between the gate insulating layer and the oxide semiconductor layer in response to the application of a positive bias voltage to the gate electrode layer, which causes the threshold voltage of the transistor to move in the negative direction. However, usage of a material with a large work function for the gate electrode layer can move the positive ions at the interface between the oxide semiconductor layer and the gate insulating layer to the gate electrode layer side.

FIG. 12 shows a schematic diagram example of a band structure of an OSFET model which includes an In—Ga—Zn—O film as an oxide semiconductor layer and also includes an In—Ga—Zn—O film including nitrogen as a gate electrode layer. Here, the In—Ga—Zn—O film, which is the oxide semiconductor layer (denoted by OS in FIG. 12), has an electron affinity of 4.6 eV and a band gap of 3.2 eV. The In—Ga—Zn—O film including nitrogen, which is the gate electrode layer (denoted by GE in FIG. 12), has a work function of 5.6 eV and a band gap of 1.8 eV. Note that in FIG. 12, the oxide semiconductor layer is n-type, and the Fermi level E_F is located above the center of the band gap.

As shown in FIG. 12, the energy band of the In—Ga—Zn—O film curves upward in the vicinity of the interface with the gate insulating layer (denoted by GI in FIG. 12), and the flat band voltage V_{FB} is higher than 0. Thus, electric field is generated in the gate insulating layer from the interface with the oxide semiconductor layer toward the interface with the gate electrode layer; accordingly, the interface with the oxide semiconductor layer is positively charged and the interface with the gate electrode layer is negatively charged. That is, the positive ions at the interface with the oxide semiconductor layer move to the gate electrode layer side.

In the above-described manner, the usage of the material with a large work function (e.g., the In—Ga—Zn—O film including nitrogen) for the gate electrode layer also has an effect of drawing positive ions at the interface with the oxide semiconductor layer to the gate electrode layer side.

With one embodiment of the present invention, a normally-off transistor or a transistor which has a threshold voltage close to 0 V although being normally-on can be achieved.

BRIEF DESCRIPTION OF DRAWINGS

In the accompanying drawings:

FIGS. 1A and 1B are a plan view and a cross-sectional view, respectively, illustrating one embodiment of a semiconductor device;

FIGS. 2A to 2F illustrate an example of a method for manufacturing a semiconductor device;

FIGS. 3A and 3B are a cross-sectional view and a circuit diagram, respectively, illustrating one embodiment of a semiconductor device;

FIGS. 4A and 4B are a circuit diagram and a perspective view, respectively, illustrating one embodiment of a semiconductor device;

FIG. 5A is a block diagram illustrating one embodiment of a semiconductor device and FIGS. 5B and 5C are circuit diagrams each illustrating part of the semiconductor device;

FIGS. 6A to 6C illustrate electronic devices;

FIGS. 7A to 7C illustrate an electronic device;

FIGS. 8A to 8C illustrate electronic devices;

FIG. 9 illustrates a transistor structure used in calculation;

FIGS. 10A to 10C show I_d - V_g characteristics obtained by calculation;

FIG. 11A is a graph showing I_d - V_g characteristics and a line obtained by second-order differentiation of the I_d - V_g characteristics and FIG. 11B shows current density distributions in a film thickness direction at several gate voltages;

FIG. 12 is a schematic view of a band structure in an OSFET model; and

FIG. 13A is a cross-sectional view and FIGS. 13B and 13C are circuit diagrams illustrating one embodiment of a semiconductor device.

BEST MODE FOR CARRYING OUT THE INVENTION

Embodiments of the present invention will be described below with reference to the drawings. Note that the present invention is not limited to the following description, and it will be easily understood by those skilled in the art that the mode and details can be changed in various ways. Therefore, the present invention should not be construed as being limited to the description in the following embodiments.

Note that in structures of the present invention described below, the same portions or portions having similar functions are denoted by the same reference numerals throughout different drawings, and description thereof is not repeated. Further, the same hatching pattern is applied to portions having similar functions, and the portions are not especially denoted by reference numerals in some cases.

Note that in each drawing in this specification, the size, the film thickness, or the region of each component may be exaggerated for clarity in some cases. Therefore, the scale is not limited to those in the drawings.

Note that in this specification and the like, the ordinal numbers such as “first” and “second” are used for convenience and do not denote the order of steps or the stacking order of layers. In addition, the ordinal numbers in this specification and the like do not denote particular names which specify the present invention.

In this specification, a term “parallel” indicates that the angle formed between two straight lines is greater than or equal to -10° and less than or equal to 10° , and accordingly also includes the case where the angle is greater than or equal to -5° and less than or equal to 5° .

In this specification, the trigonal and rhombohedral crystal systems are included in the hexagonal crystal system.

Embodiment 1

In this embodiment, one embodiment of a semiconductor device and one embodiment of a method for manufacturing the semiconductor device are described with reference to FIGS. 1A and 1B and FIGS. 2A to 2F. In this embodiment, a transistor including an oxide semiconductor stacked layers is described as an example of the semiconductor device.

FIGS. 1A and 1B illustrate an example of a structure of a transistor 120. FIG. 1A is a plan view of the transistor 120, and FIG. 1B is a cross-sectional view taken along a chain line X1-Y1 in FIG. 1A.

As illustrated in the cross-sectional view in the channel length direction in FIG. 1B, the transistor 120 includes, over a substrate 100 having an insulating surface, a gate electrode layer 102, an insulating layer 104, an oxide semiconductor stacked layers 106 including an oxide semiconductor layer 106a and an oxide semiconductor layer 106b, a source electrode layer 108a, a drain electrode layer 108b, an insulating layer 110, and a gate electrode layer 112.

In the transistor 120, the gate electrode layer 102 overlaps with the oxide semiconductor stacked layers 106 with the insulating layer 104 sandwiched therebetween, and the gate electrode layer 112 overlaps with the oxide semiconductor stacked layers 106 with the insulating layer 110 sandwiched therebetween. Further, the oxide semiconductor layer 106b is provided over and in contact with the oxide semiconductor layer 106a. The source electrode layer 108a and the drain electrode layer 108b are in contact with the oxide semiconductor layer 106b of the oxide semiconductor stacked layers 106. In addition, the insulating layer 110 is provided over and in contact with the source electrode layer 108a, the drain electrode layer 108b, and part of the oxide semiconductor stacked layers 106. Further, in the oxide semiconductor stacked layers 106, a region in contact with the insulating layer 110 has a smaller thickness than regions in contact with the source electrode layer 108a or the drain electrode layer 108b.

In the oxide semiconductor stacked layers 106, the region with a small thickness is formed by etching a part of the oxide semiconductor stacked layers 106 in processing a conductive film for forming the source electrode layer 108a and the drain electrode layer 108b or by performing etching treatment on an exposed region of the oxide semiconductor stacked layers 106 after forming the source electrode layer 108a and the drain electrode layer 108b. The region with a small thickness serves as a channel formation region of the transistor 120. In the oxide semiconductor stacked layers 106, since the channel formation region has a small thickness, the number of oxygen vacancies included in the region can be smaller than that in the other regions; thus, such a small thickness of the channel formation region can result in a reduction in the density of donors in the channel formation region.

Further, by reducing the thickness of the channel formation region in the oxide semiconductor stacked layers 106, the resistance of the regions in contact with the source electrode layer 108a or the drain electrode layer 108b can be lower than that of the channel formation region. Thus, contact resistance with the source electrode layer 108a and the drain electrode layer 108b can be reduced.

As described above, in a transistor using an oxide semiconductor layer, the decrease in the thickness of the oxide semiconductor layer, which is the active layer, and the decrease in the density of donors included in the oxide semiconductor layer are the keys to achieving a normally-off transistor. Since the transistor 120 described in this embodiment includes the oxide semiconductor stacked layers 106 with the thin channel formation region, the movement of the threshold voltage of the transistor in the negative direction can be suppressed.

In this embodiment, an example in which part of the oxide semiconductor layer 106b in the oxide semiconductor stacked layers 106 is etched to form the region with small thickness in the oxide semiconductor stacked layers 106 is

described. However, without limitation to this method, the region with a small thickness may be formed by etching part of the oxide semiconductor layer **106a** or by etching part of the oxide semiconductor layer **106a** and part of the oxide semiconductor layer **106b**, in embodiments of the present invention.

It is preferable that the oxide semiconductor layer **106b** in the oxide semiconductor stacked layers **106** include an oxide semiconductor which includes at least indium (In) and gallium (Ga) and has the following relation between the indium content and the gallium content: $\text{In} \leq \text{Ga}$. The formation energy of oxygen vacancies is larger and thus oxygen vacancies are less likely to generate in Ga than in In; therefore, the oxide having a composition relation of $\text{In} \leq \text{Ga}$ has more stable characteristics than the oxide having a composition relation of $\text{In} > \text{Ga}$. With such an oxide semiconductor material, the reliability of the transistor can be improved.

Further, it is preferable that the oxide semiconductor layer **106a** in the oxide semiconductor layers **106** include an oxide semiconductor which includes at least In and Ga and has the following relation between the indium content and the gallium content: $\text{In} > \text{Ga}$. In an oxide semiconductor, the s orbital of heavy metal mainly contributes to carrier transfer, and when the In content in the oxide semiconductor is increased, overlap of the s orbitals is likely to be increased. Therefore, the oxide having a composition relation of $\text{In} > \text{Ga}$ has higher mobility than the oxide having a composition relation of $\text{In} \leq \text{Ga}$. Accordingly, when a channel is formed in the oxide semiconductor layer **106a**, mobility of the transistor can be increased.

The thickness of the oxide semiconductor layer **106a** is preferably more than or equal to 3 nm and less than or equal to 15 nm, and further preferably more than or equal to 7 nm and less than or equal to 12 nm. In addition, the thickness of the oxide semiconductor layer **106b** in the regions in contact with the source electrode layer **108a** or the drain electrode layer **108b** is preferably more than or equal to 3 nm and less than or equal to 15 nm, and further preferably more than or equal to 5 nm and less than or equal to 12 nm. Further, the thickness of the region functioning as the channel formation region (the region between the source electrode layer **108a** and the drain electrode layer **108b**) in the oxide semiconductor stacked layers **106** is preferably more than or equal to 3 nm and less than 20 nm, and further preferably more than or equal to 5 nm and less than 15 nm.

The insulating layer **104** in contact with the oxide semiconductor layer **106a** preferably includes a region including oxygen in excess of the stoichiometric composition (hereinafter also referred to as an oxygen-excess region). By making the insulating layer **104** in contact with the oxide semiconductor layer **106a** include an oxygen-excess region, supply of oxygen to the oxide semiconductor layer **106a** is possible. Accordingly, release of oxygen from the oxide semiconductor layer **106a** can be prevented and oxygen vacancies can be compensated. Similarly, the insulating layer **110** in contact with the oxide semiconductor layer **106b** preferably includes an oxygen-excess region.

The gate electrode layer **102** at least on the side of the surface in contact with the insulating layer **104** is preferably formed using a material having a larger work function than the oxide semiconductor layer **106a**, and further preferably formed using a material having a work function that is 1 electron volt or more larger than the oxide semiconductor layer **106a**. Similarly, the gate electrode layer **112** at least on the side of the surface in contact with the insulating layer **110** is preferably formed using a material having a larger

work function than the oxide semiconductor layer **106b**, and further preferably formed using a material having a work function that is 1 electron volt or more larger than the oxide semiconductor layer **106b**. As such a material, an In—Ga—Zn—O film including nitrogen, an In—Sn—O film including nitrogen, an In—Ga—O film including nitrogen, an In—Zn—O film including nitrogen, a Sn—O film including nitrogen, an In—O film including nitrogen, or a metal nitride film (such as an indium nitride film, a zinc nitride film, a tantalum nitride film, or a tungsten nitride film) can be used, for example. These films each have a work function of 5 electron volts or more and thus can make the threshold voltage of the transistor positive. Accordingly, a normally-off switching transistor can be achieved. For example, in the case of using an In—Ga—Zn—O film including nitrogen, an In—Ga—Zn—O film including nitrogen at least at higher concentration than those of the oxide semiconductor layer **106a** and the oxide semiconductor layer **106b** is used.

In this embodiment, the gate electrode layer **102** is an electrode layer functioning as a back gate and the potential of the gate electrode layer **102** can be set as appropriate to a fixed potential, GND, or the like. By controlling the gate voltage applied to the back gate, the threshold voltage of the transistor **120** can be controlled. Thus, the transistor **120** can be normally-off.

Further, in the case of applying a negative bias voltage to the gate electrode layer **102**, positive ions (e.g., Na^+) contained as impurities in the insulating layer **104** can be moved to the gate electrode layer **102** side.

An example of a method for manufacturing the transistor **120** will be described below with reference to FIGS. 2A to 2F.

First, the gate electrode layer **102** is formed over the substrate **100** having an insulating surface.

There is no particular limitation on the substrate that can be used as the substrate **100** having an insulating surface as long as it has heat resistance enough to withstand heat treatment performed later. For example, a glass substrate of barium borosilicate glass, aluminoborosilicate glass, or the like, a ceramic substrate, a quartz substrate, or a sapphire substrate can be used. Alternatively, a single crystal semiconductor substrate or a polycrystalline semiconductor substrate made of silicon, silicon carbide, or the like, a compound semiconductor substrate made of silicon germanium or the like, an SOI substrate, or the like can be used. Furthermore, any of these substrates provided with a semiconductor element may be used as the substrate **100**.

The gate electrode layer **102** can be formed using a metal material such as molybdenum, titanium, tantalum, tungsten, aluminum, copper, chromium, neodymium, scandium, or the like, or an alloy material containing any of these metal materials as its main component. Alternatively, a semiconductor film typified by a polycrystalline silicon film doped with an impurity element such as phosphorus, or a silicide film such as a nickel silicide film may be used as the gate electrode layer **102**. The gate electrode layer **102** may have a single-layer structure or a stacked-layer structure. The gate electrode layer **102** may have a tapered shape with a taper angle of more than or equal to 30° and less than or equal to 70° for example. Here, the taper angle refers to an angle formed between a side surface of a layer having a tapered shape and a bottom surface of the layer.

The material of the gate electrode layer **102** may be a conductive material such as indium oxide-tin oxide, indium oxide containing tungsten oxide, indium zinc oxide containing tungsten oxide, indium oxide containing titanium oxide,

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indium tin oxide containing titanium oxide, indium oxide-zinc oxide, or indium tin oxide to which silicon oxide is added.

Note that the gate electrode layer **102** at least on the side of the surface that is to be in contact with the insulating layer **104** is preferably formed using a material having a larger work function than the oxide semiconductor layer provided in contact with the insulating layer **104**, and further preferably formed using a material having a work function that is 1 electron volt or more larger than the oxide semiconductor layer. As the conductive material having a large work function, a metal oxide including nitrogen can be used for example.

Next, an insulating layer **103** is formed over the gate electrode layer **102** so as to cover the gate electrode layer **102**. The insulating layer **103** can be provided to have a single-layer structure or a stacked-layer structure including a film including silicon oxide, silicon oxynitride, aluminum oxide, aluminum oxynitride, hafnium oxide, gallium oxide, zinc gallium oxide, or a mixed material of these materials by a plasma CVD method, a sputtering method, or the like.

The insulating layer **103** is a layer to be processed into the insulating layer **104** that is in contact with the oxide semiconductor stacked layers **106**. Accordingly, the insulating layer **103** preferably includes an oxygen-excess region. In order to provide the oxygen-excess region in the insulating layer **103**, the insulating layer **103** may be formed in an oxygen atmosphere, for example. Alternatively, oxygen may be introduced into the formed insulating layer **103** to provide the oxygen-excess region in the insulating layer **103**.

In this embodiment, oxygen **400** (at least including any of oxygen radicals, oxygen atoms, or oxygen ions) is introduced into the insulating layer **103** to form an oxygen-excess region **402** (see FIG. 2A). As a method for introducing oxygen, an ion implantation method, an ion doping method, a plasma immersion ion implantation method, plasma treatment, or the like may be used.

In FIG. 2A, the concentration peak position of oxygen introduced into the insulating layer **103** is indicated by dotted lines, and the region with the oxygen peak is illustrated as the oxygen-excess region **402**. Note that the depth to which oxygen is introduced and the distribution of the introduced oxygen are not limited to those in the structure illustrated in FIG. 2A.

Next, planarization treatment is performed on a surface of the insulating layer **103**; thus, the insulating layer **104** is formed (see FIG. 2B). The planarization treatment may be, but is not particularly limited to, polishing treatment (e.g., chemical mechanical polishing), dry etching treatment, plasma treatment, a combination of these treatments, or the like.

Note that in order not to remove the oxygen-excess region **402** in the planarization treatment for the insulating layer **103**, the depth to which the oxygen **400** is introduced, the removed thickness of the insulating layer **103**, and the like are adjusted as appropriate.

Further in order to reduce impurities such as hydrogen (including water, a hydroxyl group, and the like) and make an oxygen-excess state in the insulating layer, the insulating layer **104** (or the insulating layer **103** before the planarization treatment) may be subjected to heat treatment for removing hydrogen or a hydrogen compound (for dehydration or dehydrogenation).

Next, an oxide semiconductor film to be the oxide semiconductor layer **106a** and an oxide semiconductor film to be the oxide semiconductor layer **106b** are stacked over the insulating layer **104** and then processed into an island shape,

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so that the oxide semiconductor stacked layers **106** including the oxide semiconductor layer **106a** and the oxide semiconductor layer **106b** is formed (see FIG. 2C).

The oxide semiconductor layer **106a** and the oxide semiconductor layer **106b** may each have an amorphous structure or a crystal structure. In the case where the oxide semiconductor layer **106a** or **106b** has an amorphous structure, heat treatment may be performed on the oxide semiconductor stacked layers **106** in a later manufacturing step so that the oxide semiconductor layer has crystallinity. The heat treatment for crystallizing the amorphous oxide semiconductor layer is performed at a temperature higher than or equal to 250° C. and lower than or equal to 700° C., preferably higher than or equal to 400° C., further preferably higher than or equal to 500° C., still further preferably higher than or equal to 550° C. Note that the heat treatment can also serve as another heat treatment in the manufacturing process.

The oxide semiconductor films to be the oxide semiconductor layer **106a** and the oxide semiconductor layer **106b** can be formed by a sputtering method, a molecular beam epitaxy (MBE) method, a CVD method, a pulse laser deposition method, an atomic layer deposition (ALD) method, or the like as appropriate.

In the formation of the oxide semiconductor films, the hydrogen concentration in the oxide semiconductor films is preferably reduced as much as possible. In order to reduce the hydrogen concentration, for example, in the case where the oxide semiconductor films are formed by a sputtering method, a high-purity rare gas (typically argon), high-purity oxygen, or a high-purity mixed gas of a rare gas and oxygen, from which impurities such as hydrogen, water, a hydroxyl group, and hydride have been removed, is used as appropriate as an atmosphere gas supplied to a deposition chamber of a sputtering apparatus.

The oxide semiconductor films are formed in such a manner that a sputtering gas from which hydrogen and moisture have been removed is introduced into the deposition chamber while moisture remaining therein is removed, whereby the hydrogen concentration in the formed oxide semiconductor films can be reduced. In order to remove moisture remaining in the deposition chamber, an entrainment vacuum pump such as a cryopump, an ion pump, or a titanium sublimation pump is preferably used. The evacuation unit may be a turbo molecular pump provided with a cold trap. A cryopump has a high capability in removing a hydrogen molecule, a compound containing a hydrogen atom such as water (H₂O) (preferably, also a compound containing a carbon atom), and the like; therefore, the impurity concentration in the oxide semiconductor films formed in the deposition chamber which is evacuated using a cryopump can be reduced.

Further, in the case where the oxide semiconductor films are formed by a sputtering method, the relative density (the fill rate) of a metal oxide target which is used for deposition is greater than or equal to 90% and less than or equal to 100%, preferably greater than or equal to 95% and less than or equal to 99.9%. With the use of a metal oxide target with a high relative density, dense oxide semiconductor films can be deposited.

To reduce the impurity concentration in the oxide semiconductor films, it is also effective to form the oxide semiconductor films while the substrate **100** is kept at high temperature. The temperature at which the substrate **100** is heated may be higher than or equal to 150° C. and lower than or equal to 450° C.; the substrate temperature is preferably higher than or equal to 200° C. and lower than or

equal to 350° C. The crystalline oxide semiconductor films can be formed by heating the substrate at a high temperature in the deposition.

A structure of an oxide semiconductor film is described below.

An oxide semiconductor film is classified roughly into a single-crystal oxide semiconductor film and a non-single-crystal oxide semiconductor film. The non-single-crystal oxide semiconductor film includes any of an amorphous oxide semiconductor film, a microcrystalline oxide semiconductor film, a polycrystalline oxide semiconductor film, a c-axis aligned crystalline oxide semiconductor (CAAC-OS) film, and the like.

The amorphous oxide semiconductor film has disordered atomic arrangement and no crystalline component. A typical example thereof is an oxide semiconductor film in which no crystal part exists even in a microscopic region, and the whole of the film is amorphous.

The microcrystalline oxide semiconductor film includes a microcrystal (also referred to as nanocrystal) with a size greater than or equal to 1 nm and less than 10 nm, for example. Thus, the microcrystalline oxide semiconductor film has a higher degree of atomic order than the amorphous oxide semiconductor film. Hence, the density of defect states of the microcrystalline oxide semiconductor film is lower than that of the amorphous oxide semiconductor film.

The CAAC-OS film is one of oxide semiconductor films including a plurality of crystal parts, and most of each crystal part fits inside a cube whose one side is less than 100 nm. Thus, there is a case where a crystal part included in the CAAC-OS film fits a cube whose one side is less than 10 nm, less than 5 nm, or less than 3 nm. The density of defect states of the CAAC-OS film is lower than that of the microcrystalline oxide semiconductor film. The CAAC-OS film is described in detail below.

In a transmission electron microscope (TEM) image of the CAAC-OS film, a boundary between crystal parts, that is, a grain boundary is not clearly observed. Thus, in the CAAC-OS film, a reduction in electron mobility due to the grain boundary is less likely to occur.

According to the TEM image of the CAAC-OS film observed in a direction substantially parallel to a sample surface (cross-sectional TEM image), metal atoms are arranged in a layered manner in the crystal parts. Each metal atom layer has a morphology reflected by a surface over which the CAAC-OS film is formed (hereinafter, a surface over which the CAAC-OS film is formed is referred to as a formation surface) or a top surface of the CAAC-OS film, and is arranged in parallel to the formation surface or the top surface of the CAAC-OS film.

On the other hand, according to the TEM image of the CAAC-film observed in a direction substantially perpendicular to the sample surface (plan TEM image), metal atoms are arranged in a triangular or hexagonal configuration in the crystal parts. However, there is no regularity of arrangement of metal atoms between different crystal parts.

From the results of the cross-sectional TEM image and the plan TEM image, alignment is found in the crystal parts in the CAAC-OS film.

A CAAC-OS film is subjected to structural analysis with an X-ray diffraction (XRD) apparatus. For example, when the CAAC-OS film including an InGaZnO₄ crystal is analyzed by an out-of-plane method, a peak appears frequently when the diffraction angle (2θ) is around 31°. This peak is derived from the (009) plane of the InGaZnO₄ crystal, which indicates that crystals in the CAAC-OS film have c-axis alignment, and that the c-axes are aligned in a direction

substantially perpendicular to the formation surface or the top surface of the CAAC-OS film.

On the other hand, when the CAAC-OS film is analyzed by an in-plane method in which an X-ray enters a sample in a direction perpendicular to the c-axis, a peak appears frequently when 2θ is around 56°. This peak is derived from the (110) plane of the InGaZnO₄ crystal. Here, analysis (φ scan) is performed under conditions where the sample is rotated around a normal vector of a sample surface as an axis (φ axis) with 2θ fixed at around 56°. In the case where the sample is a single-crystal oxide semiconductor film of InGaZnO₄, six peaks appear. The six peaks are derived from crystal planes equivalent to the (110) plane. On the other hand, in the case of a CAAC-OS film, a peak is not clearly observed even when φ scan is performed with 2θ fixed at around 56°.

According to the above results, in the CAAC-OS film having c-axis alignment, while the directions of a-axes and b-axes are different between crystal parts, the c-axes are aligned in a direction parallel to a normal vector of a formation surface or a normal vector of a top surface. Thus, each metal atom layer arranged in a layered manner observed in the cross-sectional TEM image corresponds to a plane parallel to the a-b plane of the crystal.

Note that the crystal part is formed concurrently with deposition of the CAAC-OS film or is formed through crystallization treatment such as heat treatment. As described above, the c-axis of the crystal is aligned in a direction parallel to a normal vector of a formation surface or a normal vector of a top surface. Thus, for example, in the case where a shape of the CAAC-OS film is changed by etching or the like, the c-axis might not be necessarily parallel to a normal vector of a formation surface or a normal vector of a top surface of the CAAC-OS film.

Further, the degree of crystallinity in the CAAC-OS film is not necessarily uniform. For example, in the case where crystal growth leading to the CAAC-OS film occurs from the vicinity of the top surface of the film, the degree of the crystallinity in the vicinity of the top surface is higher than that in the vicinity of the formation surface in some cases. Further, when an impurity is added to the CAAC-OS film, the crystallinity in a region to which the impurity is added is changed, and the degree of crystallinity in the CAAC-OS film varies depends on regions.

Note that when the CAAC-OS film with an InGaZnO₄ crystal is analyzed by an out-of-plane method, a peak of 2θ may also be observed at around 36°, in addition to the peak of 2θ at around 31°. The peak of 2θ at around 36° indicates that a crystal having no c-axis alignment is included in part of the CAAC-OS film. It is preferable that in the CAAC-OS film, a peak of 2θ appear at around 31° and a peak of 2θ do not appear at around 36°.

In a transistor using the CAAC-OS film, change in electric characteristics due to irradiation with visible light or ultraviolet light is small. Thus, the transistor has high reliability.

The oxide semiconductor layer 106a and the oxide semiconductor layer 106b each may be any one of an amorphous oxide semiconductor film, a microcrystalline oxide semiconductor film, and a CAAC-OS film or a mixed film including two or more films of these. Note that the oxide semiconductor layer 106a and the oxide semiconductor layer 106b may be a stacked film including two or more films of an amorphous oxide semiconductor film, a microcrystalline oxide semiconductor film, and a CAAC-OS film, for example.

In the case where a CAAC-OS film is employed as the oxide semiconductor layer 106a and/or the oxide semicon-

ductor layer **106b**, the following methods can be used for forming the CAAC-OS film as examples. One of the methods is to form an oxide semiconductor film at a film formation temperature higher than or equal to 200° C. and lower than or equal to 450° C., thereby obtaining c-axis alignment substantially perpendicular to a surface. Another method is to form a thin oxide semiconductor film and then subject the film to heat treatment at a temperature higher than or equal to 200° C. and lower than or equal to 700° C., thereby obtaining c-axis alignment substantially perpendicular to a surface. The other method is to form a first thin oxide semiconductor film, subject the film to heat treatment at a temperature higher than or equal to 200° C. and lower than or equal to 700° C., and then form a second oxide semiconductor film, thereby obtaining c-axis alignment substantially perpendicular to a surface.

An oxide semiconductor including at least indium (In) is used for the oxide semiconductor layer **106a** and the oxide semiconductor layer **106b**. In particular, an oxide semiconductor including indium and zinc (Zn) is preferable. As a stabilizer for reducing variation in electric characteristics of a transistor using the oxide semiconductor, gallium (Ga) is preferably additionally contained. It is preferable that one or more elements selected from tin (Sn), hafnium (Hf), aluminum (Al), and zirconium (Zr) be contained as a stabilizer.

As another stabilizer, one or a plurality of kinds of lanthanoid such as lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), or lutetium (Lu) may be contained.

As the oxide semiconductor, for example, any of the following can be used: indium oxide; tin oxide; zinc oxide; a two-component metal oxide such as an In—Zn-based oxide, an In—Mg-based oxide, or an In—Ga-based oxide; a three-component metal oxide such as an In—Ga—Zn-based oxide (also referred to as IGZO), an In—Al—Zn-based oxide, an In—Sn—Zn-based oxide, an In—Hf—Zn-based oxide, an In—La—Zn-based oxide, an In—Ce—Zn-based oxide, an In—Pr—Zn-based oxide, an In—Nd—Zn-based oxide, an In—Sm—Zn-based oxide, an In—Eu—Zn-based oxide, an In—Gd—Zn-based oxide, an In—Tb—Zn-based oxide, an In—Dy—Zn-based oxide, an In—Ho—Zn-based oxide, an In—Er—Zn-based oxide, an In—Tm—Zn-based oxide, an In—Yb—Zn-based oxide, or an In—Lu—Zn-based oxide; and a four-component metal oxide such as an In—Sn—Ga—Zn-based oxide, an In—Hf—Ga—Zn-based oxide, an In—Al—Ga—Zn-based oxide, an In—Sn—Al—Zn-based oxide, an In—Sn—Hf—Zn-based oxide, or an In—Hf—Al—Zn-based oxide.

Note that here, for example, an In—Ga—Zn-based oxide means an oxide including In, Ga, and Zn as its main components, and there is no particular limitation on the ratio of In, Ga, and Zn. The In—Ga—Zn-based oxide may include a metal element other than the In, Ga, and Zn.

Alternatively, a material represented by $\text{InMO}_3(\text{ZnO})_m$ ($m>0$, m is not an integer) may be used as an oxide semiconductor. Note that M represents one or more metal elements selected from Ga, Fe, Mn, and Co. Alternatively, as the oxide semiconductor, a material represented by a chemical formula, $\text{In}_2\text{SnO}_5(\text{ZnO})_n$ ($n>0$, n is an integer) may be used.

For example, an In—Ga—Zn-based oxide with an atomic ratio where $\text{In:Ga:Zn}=1:1:1$ ($=1/3:1/3:1/3$), $\text{In:Ga:Zn}=2:2:1$ ($=2/5:2/5:1/5$), $\text{In:Ga:Zn}=3:1:2$ ($=1/2:1/6:1/3$), or any of oxides whose composition is in the neighborhood of the above compositions can be used. Alternatively, an In—Sn—

Zn-based oxide with an atomic ratio of $\text{In:Sn:Zn}=1:1:1$ ($=1/3:1/3:1/3$), $\text{In:Sn:Zn}=2:1:3$ ($=1/3:1/6:1/2$), or $\text{In:Sn:Zn}=2:1:5$ ($=1/4:1/8:5/8$), or any of oxides whose composition is in the neighborhood of the above compositions may be used.

However, an oxide semiconductor including indium that is included in a transistor is not limited to the materials given above; a material with an appropriate composition may be used for a transistor including an oxide semiconductor including indium depending on needed electrical characteristics (e.g., field-effect mobility, threshold voltage, and variation). In order to obtain the needed electrical characteristics, the carrier concentration, the impurity concentration, the defect density, the atomic ratio between a metal element and oxygen, the interatomic distance, the density, and the like are preferably set to appropriate values.

For example, high field-effect mobility can be obtained relatively easily in a transistor including an In—Sn—Zn-based oxide. Also in the case of a transistor including an In—Ga—Zn-based oxide, the field-effect mobility can be increased by reducing the defect density in a bulk.

Note that for example, the expression “the composition of an oxide including In, Ga, and Zn at the atomic ratio, $\text{In:Ga:Zn}=a:b:c$ ($a+b+c=1$), is in the neighborhood of the composition of an oxide including In, Ga, and Zn at the atomic ratio, $\text{In:Ga:Zn}=A:B:C$ ($A+B+C=1$)” means that a , b , and c satisfy the following relation: $(a-A)^2+(b-B)^2+(c-C)^2 \leq r^2$, and r may be 0.05, for example. For example, r may be 0.05. The same applies to other oxides.

Further, oxide semiconductors whose crystallinities are different from each other may be employed for the oxide semiconductor layer **106a** and the oxide semiconductor layer **106b**. That is, an appropriate combination of any of a single crystal oxide semiconductor, a polycrystalline oxide semiconductor, a microcrystalline oxide semiconductor, an amorphous oxide semiconductor, and a CAAC-OS may be employed.

Further, the oxide semiconductor stacked layers **106** is preferably subjected to heat treatment for removing excess hydrogen (including water and a hydroxyl group) contained in the oxide semiconductor stacked layers **106** (for dehydration or dehydrogenation). The temperature of the heat treatment is higher than or equal to 300° C. and lower than or equal to 700° C., or lower than the strain point of the substrate. The heat treatment can be performed under reduced pressure, a nitrogen atmosphere, or the like. Hydrogen, which is an impurity imparting n-type conductivity, can be removed from the oxide semiconductor by this heat treatment.

Note that the heat treatment for the dehydration or dehydrogenation may be performed at any timing in the manufacturing process of the transistor **120** as long as the heat treatment is performed after the formation of the oxide semiconductor layer. The heat treatment for dehydration or dehydrogenation may be performed a plurality of times, or may also serve as another heat treatment.

Note that the heat treatment for dehydration or dehydrogenation is preferably performed before the oxide semiconductor films are processed into an island shape because oxygen contained in the insulating layer **104** can be prevented from being released by the heat treatment.

In the heat treatment, it is preferable that water, hydrogen, or the like be not contained in nitrogen or a rare gas such as helium, neon, or argon. The purity of nitrogen or the rare gas such as helium, neon, or argon which is introduced into the heat treatment apparatus is set to preferably 6N (99.9999%) or higher, further preferably 7N (99.99999%) or higher (that

is, the impurity concentration is preferably 1 ppm or lower, further preferably 0.1 ppm or lower).

In addition, after the oxide semiconductor stacked layers **106** (or the oxide semiconductor films before the processing into the island shape) is heated by the heat treatment, a high-purity oxygen gas, a high-purity dinitrogen gas, or ultra dry air (the moisture amount is less than or equal to 20 ppm (-55°C . by conversion into a dew point), preferably less than or equal to 1 ppm, further preferably less than or equal to 10 ppb, in the measurement with use of a dew point meter of a cavity ring down laser spectroscopy (CRDS) system) may be introduced into the same furnace while the heating temperature is being maintained or being gradually decreased. It is preferable that water, hydrogen, or the like be not contained in the oxygen gas or the dinitrogen monoxide gas. The purity of the oxygen gas or the dinitrogen monoxide gas which is introduced into the heat treatment apparatus is preferably 6N or more, further preferably 7N or more (i.e., the impurity concentration in the oxygen gas or the dinitrogen monoxide gas is preferably 1 ppm or lower, further preferably 0.1 ppm or lower). The oxygen gas or the dinitrogen monoxide gas acts to supply oxygen that is a main component of the oxide semiconductor to compensate for a reduction of oxygen in the step for removing an impurity for the dehydration or dehydrogenation, so that the oxide semiconductor layers can be highly-purified and i-type (intrinsic) oxide semiconductor layers.

Since there is a possibility that oxygen, which is a main component of an oxide semiconductor, is also released and reduced by dehydration or dehydrogenation treatment, oxygen (including at least one of an oxygen radical, an oxygen atom, and an oxygen ion) may be introduced into the oxide semiconductor layer which has been subjected to the dehydration or dehydrogenation treatment to supply oxygen to the layer.

Oxygen is added and supplied to the dehydrated or dehydrogenated oxide semiconductor layer, so that the oxide semiconductor layer can be highly purified and be i-type (intrinsic). Variation in electric characteristics of a transistor having the highly-purified and i-type (intrinsic) oxide semiconductor is suppressed, and the transistor is electrically stable.

In the step of addition of oxygen to the oxide semiconductor layer, oxygen may be directly added to the oxide semiconductor layer or may be added to the oxide semiconductor layer through another film such as the insulating layer **110** to be formed later. When oxygen is introduced through another film, an ion implantation method, an ion doping method, a plasma immersion ion implantation method, or the like may be used. In the case where oxygen is directly introduced into the exposed oxide semiconductor layer, plasma treatment or the like can be used in addition to the above-described methods.

As a gas for supplying oxygen, a gas containing O may be used; for example, an O_2 gas, an N_2O gas, a CO_2 gas, a CO gas, or an NO_2 gas may be used. Note that a rare gas (e.g., an Ar gas) may be contained in the supply gas of the oxygen.

For example, in the case where an oxygen ion is introduced into the oxide semiconductor layer by an ion implantation method, the dose can be greater than or equal to 1×10^{13} ions/ cm^2 and less than or equal to 5×10^{16} ions/ cm^2 .

Alternatively, oxygen may be supplied to the oxide semiconductor layer in the following manner: an insulating layer in contact with the oxide semiconductor layer (the insulating layer **104** or the insulating layer **110**) is formed to have an oxygen-excess region; and heat treatment is performed in a state where the insulating layer and the oxide semiconductor

layer are in contact with each other, so that excess oxygen contained in the insulating layer is diffused to the oxide semiconductor layer. This heat treatment can serve as another heat treatment in the process for manufacturing the transistor **120**.

The timing of supply of oxygen to the oxide semiconductor layer is not particularly limited to the above as long as it is after the formation of the oxide semiconductor layer. The step of introducing oxygen into the oxide semiconductor layer may be performed a plurality of times. Further, heat treatment for dehydration or dehydrogenation and/or supply of oxygen may be performed separately on each of the oxide semiconductor layers or may be performed once on the oxide semiconductor stacked layers after the oxide semiconductor stacked layers **106** having a stacked-layer structure is formed.

The insulating layer **104** and the oxide semiconductor film to be processed into the oxide semiconductor layer **106a** are preferably formed in succession without being exposed to air. By the successive formation of the insulating layer **104** and the oxide semiconductor film, impurities such as hydrogen and moisture can be prevented from being adsorbed onto a surface of the insulating layer **104**.

Next, a conductive film **108** to be processed into the source electrode layer **108a** and the drain electrode layer **108b** is formed over the oxide semiconductor stacked layers **106** (see FIG. 2D). As the conductive film **108**, for example, a metal film containing an element selected from Al, Cr, Cu, Ta, Ti, Mo, or W, or a metal nitride film containing any of the above elements as a component (e.g., a titanium nitride film, a molybdenum nitride film, or a tungsten nitride film) can be used. Alternatively, a film of a high-melting-point metal such as Ti, Mo, or W or a metal nitride film of any of these elements (a titanium nitride film, a molybdenum nitride film, or a tungsten nitride film) may be stacked on one of or both a bottom side and a top side of a metal film of Al, Cu, or the like. Further alternatively, the conductive film used for the source electrode layer **108a** and the drain electrode layer **108b** may be formed using a conductive metal oxide. As the conductive metal oxide, indium oxide (In_2O_3), tin oxide (SnO_2), zinc oxide (ZnO), indium oxide-tin oxide alloy ($\text{In}_2\text{O}_3\text{—SnO}_2$), an indium oxide-zinc oxide alloy ($\text{In}_2\text{O}_3\text{—ZnO}$), or any of these metal oxide materials in which silicon oxide is contained can be used.

As the conductive film **108**, a metal nitride film such as an In—Ga—Zn—O film including nitrogen, an In—Sn—O film including nitrogen, an In—Ga—O film including nitrogen, an In—Zn—O film including nitrogen, a Sn—O film including nitrogen, or an In—O film including nitrogen can be used. These films include the same constituent elements as the oxide semiconductor layer **106b** and can therefore stabilize the interface with the oxide semiconductor layer **106b**. For example, the conductive film **108** can have a stacked-layer structure in which an In—Ga—Zn—O film including nitrogen and a tungsten film are stacked from the side in contact with the oxide semiconductor layer **106b**.

Next, the conductive film **108** is selectively etched to form the source electrode layer **108a** and the drain electrode layer **108b** (see FIG. 2E). The source electrode layer **108a** and the drain electrode layer **108b** may have a tapered shape. It is preferable that the source electrode layer **108a** and the drain electrode layer **108b** have a tapered shape to relieve concentration of electric field between the source and the drain.

By this etching treatment, part of the oxide semiconductor layer **106b** is also etched, so that the region with a small thickness is formed between the source electrode layer **108a** and the drain electrode layer **108b**. Alternatively, after the

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source electrode layer **108a** and the drain electrode layer **108b** are formed, the exposed part of the oxide semiconductor layer **106b** may be subjected to etching treatment (e.g., wet etching treatment) to form the region with a small thickness. The thickness of the oxide semiconductor layer **106b** in the region with a small thickness, which is formed by processing treatment on the source electrode layer **108a** and the drain electrode layer **108b** and/or later etching treatment, is preferably more than or equal to 1 nm and less than or equal to 2 nm.

Then, the insulating layer **110** is formed to cover the exposed oxide semiconductor layer **106b**, the source electrode layer **108a**, and the drain electrode layer **108b**. The insulating layer **110** functions as a gate insulating layer. A conductive film to be the gate electrode layer **112** (including a wiring formed of the same film) is formed over the insulating layer **110**, and is selectively etched, so that the gate electrode layer **112** is formed (see FIG. 2F).

The insulating layer **110** can be formed using a similar material and formation method to those of the insulating layer **104**. Note that the thickness of the insulating layer **110** is preferably more than or equal to 5 nm and less than or equal to 30 nm, further preferably more than or equal to 9 nm and less than or equal to 22 nm.

The gate electrode layer **112** can be formed using similar material and formation method to those of the gate electrode layer **102**. Note that the gate electrode layer **112** at least on the side of the surface that is in contact with the insulating layer **110** is preferably formed using a material having a larger work function than the oxide semiconductor layer **106b**, and further preferably formed using a material having a work function that is 1 electron volt or more larger than the oxide semiconductor layer **106b**.

In the above-described manner, the transistor **120** in this embodiment can be manufactured.

The transistor **120** described in this embodiment includes an oxide semiconductor stacked layers **106** in which a channel formation region is thinner than the other region (e.g., a region in contact with the source electrode layer **108a** or the drain electrode layer **108b** and excluding a tapered portion at the edge of the oxide semiconductor layer). This can suppress the shift of the threshold voltage of the transistor **120** in the negative direction.

Further, the transistor **120** described in this embodiment includes the gate electrode layer **102** and the gate electrode layer **112** between which the semiconductor stacked layers **106** including a channel formation region is sandwiched. By application of a negative bias voltage to the gate electrode layer **102**, generation of the second current on the back channel side can be suppressed. In this way, the threshold voltage of the transistor **120** can be moved in the positive direction.

The structures, methods, and the like which are described in this embodiment can be combined as appropriate with any of the structures, methods, and the like which are described in the other embodiments.

Embodiment 2

In this embodiment, as an example of a semiconductor device which includes the transistor described in Embodiment 1, a semiconductor device which can hold stored data even when not powered and which has an unlimited number of write cycles will be described with reference to drawings.

FIGS. 3A and 3B illustrate one example of a structure of the semiconductor device. FIG. 3A is a cross-sectional view

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of the semiconductor device, and FIG. 3B is a circuit diagram of the semiconductor device.

The semiconductor device illustrated in FIG. 3A includes a transistor **160** including a first semiconductor material in a lower portion, and a transistor **162** including a second semiconductor material in an upper portion. As the transistor **162**, the transistor of one embodiment of the present invention described in Embodiment 1 can be employed.

Here, the first semiconductor material and the second semiconductor material are preferably materials having different band gaps. For example, the first semiconductor material may be a semiconductor material (e.g., silicon) other than an oxide semiconductor, and the second semiconductor material may be an oxide semiconductor. A transistor including a material other than an oxide semiconductor can operate at high speed easily. On the other hand, a transistor including an oxide semiconductor has characteristics of enabling holding of charge for a long time.

Although the above-described transistors are both n-channel transistors here, it is needless to say that p-channel transistors can be used. The specific structure of the semiconductor device, such as the material used for the semiconductor device and the structure of the semiconductor device, is not necessarily limited to those described here except for the use of the transistor described in Embodiment 1, which is formed using an oxide semiconductor to hold data, as the transistor **162**.

The transistor **160** in FIG. 3A includes a channel formation region **216** provided in a substrate **200** including a semiconductor material (e.g., silicon), impurity regions **214** and high-concentration impurity regions **220** (these are simply collectively referred to as impurity regions) which are provided so that the channel formation region **216** is sandwiched therebetween, intermetallic compound regions **224** in contact with the high-concentration impurity regions **220**, a gate insulating layer **208** provided over the channel formation region **216**, a gate electrode layer **210** provided over the gate insulating layer **208**, sidewall insulating layers **218** provided on side surfaces of the gate electrode layer **210**, an electrode layer **212a**, and an electrode layer **212b**.

The electrode layer **212a** and the electrode layer **212b** are electrode layers functioning as a source electrode layer and a drain electrode layer and are electrically connected to the intermetallic compound regions **224** through contact holes provided in the insulating layer **228** that is formed over the gate electrode layer **210**. The insulating layer **228** may have a single-layer structure or a stacked-layer structure including an inorganic insulating film such as a silicon oxide film, a silicon oxynitride film, an aluminum oxide film, an aluminum oxynitride film, a silicon nitride film, an aluminum nitride film, a silicon nitride oxide film, or an aluminum nitride oxide film.

An element isolation insulating layer **206** is provided over the substrate **200** to surround the transistor **160**.

The transistor **160** formed using a single crystal semiconductor substrate can operate at high speed. Thus, when the transistor is used as a reading transistor, data can be read at a high speed.

The transistor **162** illustrated in FIG. 3A is a transistor in which an oxide semiconductor is used for a channel formation region. A transistor in which an oxide semiconductor is used for a channel formation region can achieve extremely small off-state current characteristics. Note that an oxide semiconductor layer included in the transistor **162** is preferably highly purified. By using a highly purified oxide semiconductor, the transistor **162** can have more favorable off-state current characteristics.

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Since the off-state current of the transistor **162** is small, stored data can be held for a long time owing to such a transistor. In other words, the semiconductor memory device in which refresh operation is unnecessary or the frequency of refresh operation is extremely low can be provided, which leads to a sufficient reduction of power consumption.

The transistor **162** includes an oxide semiconductor stacked layers **244** which includes an oxide semiconductor layer **244a** and an oxide semiconductor layer **244b** and in which a channel formation region is thinner than the other region (e.g., a region in contact with an electrode layer **268a** or an electrode layer **268b**). In the oxide semiconductor stacked layers **244** included in the transistor **162**, the channel formation region is thinned, which can suppress the shift of the threshold voltage in the negative direction. Further, the transistor **162** includes a gate electrode layer **202b** which overlaps with the oxide semiconductor stacked layers **244** with an insulating layer **203** and an insulating layer **204** sandwiched therebetween, in addition to a gate electrode layer **262** which overlaps with the oxide semiconductor stacked layers **244** with a gate insulating layer **260** sandwiched therebetween. The gate electrode layer **202b** can be used as a back gate electrode. By application of a negative bias voltage to the gate electrode layer **202b**, flow of the second current on the back channel side can be suppressed, which can move the threshold voltage of the transistor **162** in the positive direction. Consequently, the transistor **162** can be a normally-off transistor.

The insulating layer **203** and the insulating layer **204** can each be a film including silicon oxide, silicon oxynitride, aluminum oxide, aluminum oxynitride, hafnium oxide, gallium oxide, zinc gallium oxide, or a mixed material of these materials. Note that the insulating layer **204** in contact with the oxide semiconductor stacked layers **244** (specifically the oxide semiconductor layer **244a**) preferably includes an oxygen-excess region.

Note that it is preferable that the insulating layer **203** be a film having a barrier property against oxygen, in order to prevent release of oxygen from the insulating layer **204**. As the film having a barrier property against oxygen, a film having a lower oxygen-transmitting property than the insulating layer **204** can be used, and specifically a single-layer structure or a stacked-layer structure including an oxide film or a nitride film of, for example, aluminum, aluminum to which magnesium is added, aluminum to which titanium is added, magnesium, titanium, or the like can be employed. Furthermore, as the insulating layer **203**, a film having a low impurity (e.g., hydrogen, moisture)-transmitting property in addition to having a barrier property against oxygen is preferably used. As such a film, an aluminum oxide film can preferably be used. With the use of an aluminum oxide film as the insulating layer **203**, release of oxygen is prevented, and in addition, entry of impurities such as hydrogen and moisture, which might cause variation in electric characteristics of the transistor **162**, can be suppressed.

Note that for favorable coverage by the insulating layer **203**, it is preferable that the gate electrode layer **202b** and an electrode layer **202a** have a tapered shape. The taper angle is preferably more than or equal to 30° and less than or equal to 70°.

An insulating layer **232** and an insulating layer **236** with a single-layer structure or a stacked-layer structure are provided over the transistor **162**. As the insulating layer **232** or the insulating layer **236**, a film including a material similar to that of the insulating layer **203** and the insulating layer **204** can be used. If necessary, planarization treatment such as CMP treatment may be performed after the insulat-

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ing layer **236** is formed to planarize a surface of the insulating layer **236**. Alternatively, as the insulating layer **236**, a planarization insulating film may be formed in order to reduce surface roughness due to the transistor or an inorganic insulating film and a planarization insulating film may be stacked. For the planarization insulating film, an organic material such as a polyimide-based resin, an acrylic-based resin, or a benzocyclobutene-based resin can be used. Besides the above organic materials, a low-dielectric constant material (a low-k material) or the like can be used.

A wiring layer **256** is provided over the insulating layer **236**. The wiring layer **256** is a wiring which connects the transistor **162** to another transistor. The wiring layer **256** is electrically connected to the electrode layer **268b** through a contact hole formed in the insulating layer **236**, the insulating layer **232**, and the gate insulating layer **260**. Note that an electrode layer may be additionally formed in the contact hole to electrically connect the wiring layer **256** to the electrode layer **268b**.

In addition, a conductive layer **253** is provided in a region overlapping with the electrode layer **268a** of the transistor **162** with the gate insulating layer **260** sandwiched therebetween. The electrode layer **268a**, the gate insulating layer **260**, and the conductive layer **253** form a capacitor **164**. That is, the electrode layer **268a** of the transistor **162** functions as one electrode of the capacitor **164**, and the conductive layer **253** functions as the other electrode of the capacitor **164**. Note that the capacitor **164** may be omitted if a capacitor is not needed. Alternatively, the capacitor **164** may be independently provided above the transistor **162**.

In this embodiment, the conductive layer **253** can be formed in the same manufacturing step as the gate electrode layer **262** of the transistor **162**.

The electrode layer **268a** is electrically connected to the electrode layer **202a** that is formed in the same layer as the gate electrode layer **202b**. Further, the electrode layer **202a** is electrically connected to an electrode layer **222a** through a contact hole provided in an insulating layer **234**. Although not shown in FIG. 3A, the electrode layer **222a** is electrically connected to the gate electrode layer **210** of the transistor **160**. Accordingly, the electrode layer **268a** of the transistor **162** is electrically connected to the gate electrode layer **210** of the transistor **160**.

An insulating layer **230** and the insulating layer **234** can each have a structure similar to that of the insulating layer **228**. Note that if necessary, the insulating layer **228**, the insulating layer **230**, and the insulating layer **234** may be subjected to planarization treatment. Moreover, the structure enabling electrical connection between the electrode layer **268a** of the transistor **162** and the gate electrode layer **210** of the transistor **160** is not limited to the structure illustrated in FIG. 3A, and the structure regarding intervening electrode layers (or wiring layers) or insulating layers can be determined as appropriate. For example, an electrode layer may be additionally provided between the electrode layer **202a** and the electrode layer **222a**, or the electrode layer **268a** may be directly connected to the gate electrode layer **210**.

Note that in the case where the insulating layer **204** includes an oxygen-excess region, there is a possibility that excess oxygen included in the insulating layer **204** may be released at the time of forming the contact hole; for this reason, the contact hole is preferably provided in a region not overlapping with the oxide semiconductor stacked layers **244**. In FIG. 3A, the electrode layer **202a** is electrically connected to the electrode layer **268a** through the contact hole provided in the insulating layer **204** in the region not overlapping with the oxide semiconductor stacked layers

244. Note that a contact hole provided below the insulating layer 204 (on the transistor 160 side) may overlap with the oxide semiconductor stacked layers 244. In the example illustrated in FIG. 3A, the gate electrode layer 202b of the transistor 162 is electrically connected to the wiring layer 222b provided in the same layer as the electrode layer 222a.

In FIG. 3A, the transistor 160 and the transistor 162 are provided so as to at least partly overlap with each other. Further, the transistor 162 and the capacitor 164 are preferably provided so as to overlap with at least part of the transistor 160. For example, the conductive layer 253 included in the capacitor 164 is provided so as to at least partly overlap with the gate electrode layer 210 of the transistor 160. With such a planar layout, the area occupied by the semiconductor device can be reduced; thus, higher integration can be achieved.

Next, an example of a circuit configuration corresponding to FIG. 3A is illustrated in FIG. 3B.

In FIG. 3B, a first wiring (1st Line) is electrically connected to the source electrode layer of the transistor 160. A second wiring (2nd Line) is electrically connected to the drain electrode layer of the transistor 160. A third wiring (3rd Line) is electrically connected to one of a source electrode layer and a drain electrode layer of the transistor 162, and a fourth wiring (4th Line) is electrically connected to a gate electrode layer of the transistor 162. The gate electrode layer of the transistor 160 and the other of the source electrode layer and the drain electrode layer of the transistor 162 are electrically connected to one electrode of the capacitor 164. A fifth wiring (5th line) and the other electrode of the capacitor 164 are electrically connected to each other.

The semiconductor device illustrated in FIG. 3B utilizes a characteristic of capable of holding the potential of the gate electrode layer of the transistor 160, and thus enables data writing, holding, and reading as follows.

Writing and holding of data will be described. First, the potential of the fourth line is set to a potential at which the transistor 162 is turned on, so that the transistor 162 is turned on. Accordingly, the potential of the third wiring is supplied to the gate electrode layer of the transistor 160 and to the capacitor 164. That is, predetermined charge is supplied to the gate electrode layer of the transistor 160 (writing). Here, one of two kinds of charges providing different potential levels (hereinafter referred to as low-level charge and high-level charge) is given. After that, the potential of the fourth wiring is set to a potential at which the transistor 162 is turned off, so that the transistor 162 is turned off. Thus, the charge given to the gate electrode layer of the transistor 160 is held (holding).

Since the amount of off-state current of the transistor 162 is extremely small, the charge of the gate electrode layer of the transistor 160 is held for a long time.

Next, reading of data will be described. By supplying an appropriate potential (a reading potential) to the fifth wiring while supplying a predetermined potential (a constant potential) to the first wiring, the potential of the second wiring varies depending on the amount of charge held at the gate electrode layer of the transistor 160. This is because in general, when the transistor 160 is an n-channel transistor, an apparent threshold voltage V_{th_H} in the case where the high-level charge is given to the gate electrode layer of the transistor 160 is lower than an apparent threshold voltage V_{th_L} in the case where the low-level charge is given to the gate electrode layer of the transistor 160. Here, an apparent threshold voltage refers to the potential of the fifth line, which is needed to turn on the transistor 160. Thus, the

potential of the fifth wiring is set to a potential V_0 that is between V_{th_H} and V_{th_L} , whereby charge supplied to the gate electrode layer of the transistor 160 can be determined. For example, in the case where a high-level charge is given in writing, when the potential of the fifth wiring is set to V_0 ($>V_{th_H}$), the transistor 160 is turned on. In the case where a low-level charge is given in writing, even when the potential of the fifth wiring is set to V_0 ($<V_{th_L}$), the transistor 160 remains in an off state. Therefore, the stored data can be read by the potential of the second line.

Note that in the case where memory cells are arrayed to be used, only data of desired memory cells needs to be read. In the case where data is not read out, a potential at which the transistor 160 is off regardless of the state of the gate electrode layer of the transistor 160, that is, a potential smaller than V_{th_H} may be given to the fifth wiring. Alternatively, a potential which allows the transistor 160 to be on regardless of a state of the gate electrode layer, that is, a potential higher than V_{th_L} may be given to the fifth wiring.

When a transistor having a channel formation region formed using an oxide semiconductor and having extremely small off-state current is applied to the semiconductor device in this embodiment, the semiconductor device can store data for an extremely long period. In other words, power consumption can be adequately reduced because refresh operation becomes unnecessary or the frequency of refresh operation can be extremely low. Moreover, stored data can be held for a long period even when power is not supplied (note that a potential is preferably fixed). Here, by achieving normally-off characteristics of the transistor 162, a configuration in which a ground potential can be input to the gate (gate electrode layer 262) of the transistor 162 when power is off can be made. With this configuration, the transistor 162 can remain off and stored data can be kept held even when power is off.

Further, in the semiconductor device described in this embodiment, high voltage is not needed for writing data and there is no problem of deterioration of elements. For example, unlike a conventional nonvolatile memory, it is not necessary to inject and extract electrons into and from a floating gate, and thus a problem such as deterioration of a gate insulating layer does not occur at all. In other words, the semiconductor device according to one embodiment of the present invention does not have a limit on the number of write cycles which is a problem in a conventional nonvolatile memory, and reliability thereof is drastically improved. Furthermore, data is written depending on the on state and the off state of the transistor, whereby high-speed operation can be easily achieved.

In the above-described manner, a miniaturized and highly-integrated semiconductor device having high electrical characteristics and a method for manufacturing the semiconductor device can be provided.

The structures, methods, and the like which are described in this embodiment can be combined as appropriate with any of the structures, methods, and the like which are described in the other embodiments.

Embodiment 3

In this embodiment, a semiconductor device including the transistor described in Embodiment 1, which can hold stored data even when not powered, which does not have a limitation on the number of write cycles, and which has a structure different from the structure described in Embodiment 2 will be described with reference to FIGS. 4A and 4B.

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FIG. 4A illustrates an example of a circuit configuration of the semiconductor device, and FIG. 4B is a conceptual diagram illustrating an example of the semiconductor device. First, the semiconductor device illustrated in FIG. 4A will be described, and then, the semiconductor device illustrated in FIG. 4B will be described.

In the semiconductor device illustrated in FIG. 4A, a bit line BL is electrically connected to the source electrode layer or the drain electrode layer of the transistor 162, a word line WL is electrically connected to the gate electrode layer of the transistor 162, and the source electrode layer or the drain electrode layer of the transistor 162 is electrically connected to a first terminal of a capacitor 254.

Next, writing and holding of data in the semiconductor device (a memory cell 250) illustrated in FIG. 4A will be described.

First, the potential of the word line WL is set to a potential at which the transistor 162 is turned on, so that the transistor 162 is turned on. Accordingly, the potential of the bit line BL is supplied to the first terminal of the capacitor 254 (writing). After that, the potential of the word line WL is set to a potential at which the transistor 162 is turned off, so that the transistor 162 is turned off. Thus, the potential of the first terminal of the capacitor 254 is held (holding).

The transistor 162 including an oxide semiconductor has extremely low off-state current. For that reason, the potential of the first terminal of the capacitor 254 (or a charge accumulated in the capacitor 254) can be held for an extremely long period by turning off the transistor 162. Here, by achieving normally-off characteristics of the transistor 162, a configuration in which a ground potential can be input to the gate of the transistor 162 when power is off can be made. With this configuration, the transistor 162 can remain off and stored data can be kept held even when power is off.

Next, reading of data will be described. When the transistor 162 is turned on, the bit line BL which is in a floating state and the capacitor 254 are electrically connected to each other, and the charge is redistributed between the bit line BL and the capacitor 254. As a result, the potential of the bit line BL is changed. The amount of change in potential of the bit line BL varies depending on the potential of the first terminal of the capacitor 254 (or the charge accumulated in the capacitor 254).

For example, the potential of the bit line BL after charge redistribution is $(C_B \times V_{B0} + C \times V)/(C_B + C)$, where V is the potential of the first terminal of the capacitor 254, C is the capacitance of the capacitor 254, C_B is the capacitance of the bit line BL (hereinafter also referred to as bit line capacitance), and V_{B0} is the potential of the bit line BL before the charge redistribution. Therefore, it can be found that assuming that the memory cell 250 is in either of two states which are the state in which the potential of the first terminal of the capacitor 254 is V_1 and the state in which the potential of the first terminal of the capacitor 254 is V_0 ($V_1 > V_0$), the potential of the bit line BL in the case of holding the potential V_1 ($= (C_B \times V_{B0} + C \times V_1)/(C_B + C)$) is higher than the potential of the bit line BL in the case of holding the potential V_0 ($= (C_B \times V_{B0} + C \times V_0)/(C_B + C)$).

Then, by comparing the potential of the bit line BL with a predetermined potential, data can be read.

As described above, the semiconductor device illustrated in FIG. 4A can hold charge that is accumulated in the capacitor 254 for a long time because the off-state current of the transistor 162 is extremely low. In other words, power consumption can be sufficiently reduced because refresh operation becomes unnecessary or the frequency of refresh

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operation can be extremely low. Moreover, stored data can be held for a long period even when power is not supplied.

Next, the semiconductor device illustrated in FIG. 4B will be described.

The semiconductor device illustrated in FIG. 4B includes a memory cell array 251a and a memory cell array 251b including a plurality of memory cells 250 illustrated in FIG. 4A as memory circuits in the upper portion, and a peripheral circuit 258 in the lower portion, which is necessary for operating the memory cell array 251 (the memory cell arrays 251a and 251b). Note that the peripheral circuit 258 is electrically connected to the memory cell array 251.

In the structure illustrated in FIG. 4B, the peripheral circuit 258 can be provided under the memory cell array 251 (the memory cell arrays 251a and 251b). Thus, the size of the semiconductor device can be decreased.

It is preferable that a semiconductor material of the transistor provided in the peripheral circuit 253 be different from that of the transistor 162. For example, silicon, germanium, silicon germanium, silicon carbide, or gallium arsenide can be used, and a single crystal semiconductor is preferably used. Alternatively, an organic semiconductor material or the like may be used. A transistor including such a semiconductor material can operate at sufficiently high speed. Further, with the transistor, a variety of circuits (such as a logic circuit or a driver circuit) which are required to operate at high speed can be achieved favorably.

Note that FIG. 4B illustrates, as an example, the semiconductor device in which two memory cell arrays 251 (the memory cell arrays 251a and 251b) are stacked; however, the number of memory cell arrays to be stacked is not limited thereto. Three or more memory cell arrays may be stacked.

In the above-described manner, a miniaturized and highly-integrated semiconductor device having high electrical characteristics and a method for manufacturing the semiconductor device can be provided.

This embodiment can be implemented in appropriate combination with any of the structures described in the other embodiments.

Embodiment 4

In this embodiment, as other examples of the semiconductor device that uses the transistor described in this specification, a NOR circuit and a NAND circuit, which are logic circuits, are illustrated in FIGS. 13A to 13C. FIG. 13B illustrates the NOR circuit, and FIG. 13C illustrates the NAND circuit. FIG. 13A is a cross-sectional view illustrating the structure of a transistor 802 and a transistor 803 in the NOR circuit in FIG. 13B.

In the NOR circuit and the NAND circuit illustrated in FIGS. 13B and 13C, p-channel transistors 801, 802, 811, and 814 can have the same structure as the transistor 160 described in Embodiment 2. In this embodiment, boron (B), aluminum (Al), gallium (Ga), or the like which is an impurity element imparting p-type conductivity is introduced into a substrate 800 that uses an n-type semiconductor material (e.g., n-type single crystal silicon substrate), to form the p-channel transistors including p-type impurity regions.

In addition, transistors each having the same structure as the transistor 120 in Embodiment 1 and including an oxide semiconductor film in which a channel formation region is formed are applied to n-channel transistors 803, 804, 812, and 813.

In the NOR circuit and the NAND circuit illustrated in FIGS. 13A to 13C, the transistors **803**, **804**, **812**, and **813** each include a thinned channel formation region in the oxide semiconductor stacked layers; accordingly, the shift of the threshold voltages of the transistors in the negative direction can be suppressed. Further, a first gate electrode layer and a second gate electrode layer are provided so that an oxide semiconductor stacked layers is sandwiched therebetween with an insulating layer sandwiched between the oxide semiconductor stacked layers and each of the first gate electrode layer and the second gate electrode. One of the gate electrode layers is used as a back gate. By controlling the potential of the back gate as appropriate, for example, to be GND, the threshold voltages of the transistors **803**, **804**, **812**, and **813** can be moved in the positive direction; consequently, the transistors can be normally-off.

In the example described in this embodiment, the gate electrode layers which are provided in the transistors **803** and the transistor **804** and each function as a back gate are electrically connected to each other in the NOR circuit, and the gate electrode layers which are provided in the transistor **812** and the transistor **813** and each function as a back gate are electrically connected to each other in the NAND circuit. However, without limitation to the above structure, a structure in which each of the gate electrode layers functioning as back gates is independently electrically controlled may be employed.

The semiconductor device illustrated in FIG. 13A is an example in which a single crystal silicon substrate is used as the substrate **800**, the transistor **802** is formed using the single crystal silicon substrate, and the transistor **803** including a channel formation region formed using the oxide semiconductor stacked layers is stacked over the transistor **802**. Over the substrate **800**, an element isolation insulating layer **806** is provided to surround the transistor **802**.

An electrode layer **841b** electrically connected to a gate electrode layer **841a** of the transistor **803** is electrically connected to an electrode layer **835** that is an electrode layer provided in the same layer as a gate electrode layer **840** through a contact hole provided in a gate insulating layer **843** and an insulating layer **839**. The electrode layer **835** is electrically connected to a wiring layer **832** through a contact hole provided in an insulating layer **836** and an insulating layer **833**. Although not clearly illustrated in FIG. 13A, the wiring layer **832** is electrically connected to a gate electrode layer **821** of the transistor **802** through a contact hole provided in an insulating layer **830** and an insulating layer **826**. Accordingly, the gate electrode layer **841a** of the transistor **803** is electrically connected to the gate electrode layer **821** of the transistor **802**.

Furthermore, although not clearly illustrated in FIG. 13A, an electrode layer **825** of the transistor **802** is electrically connected to a wiring layer **834**, and the wiring layer **834** is electrically connected to an electrode layer **845** of the transistor **803** with an electrode layer **831**. Accordingly, the electrode layer **825** of the transistor **802** is electrically connected to the electrode layer **845** of the transistor **803**.

Note that the structure enabling electrical connection between the electrode layer (or the gate electrode layer) of the transistor **802** and the electrode layer (or the gate electrode layer) of the transistor **803** is not limited to the structure illustrated in FIG. 13A, and the structure regarding intervening electrode layers (or wiring layers) or insulating layers can be determined as appropriate.

The overlapping structure of the transistor **802** and the transistor **803** illustrated in FIG. 13A can reduce the area of the semiconductor device, leading to higher integration.

Further, since the transistor **802** is a transistor that can be normally-off, the logic circuit can be controlled precisely.

In the above-described manner, a miniaturized and highly-integrated semiconductor device having high electrical characteristics and a method for manufacturing the semiconductor device can be provided.

The structures, methods, and the like which are described in this embodiment can be combined as appropriate with any of the structures, methods, and the like which are described in the other embodiments.

Embodiment 5

In this embodiment, a central processing unit (CPU) at least part of which includes the transistor disclosed above in Embodiment 1 will be described as an example of a semiconductor device.

FIG. 5A is a block diagram illustrating a specific configuration of a CPU. The CPU illustrated in FIG. 5A includes an arithmetic logic unit (ALU) **1191**, an ALU controller **1192**, an instruction decoder **1193**, an interrupt controller **1194**, a timing controller **1195**, a register **1196**, a register controller **1197**, a bus interface (Bus I/F) **1198**, a rewritable ROM **1199**, and a ROM interface (ROM I/F) **1189** over a substrate **1190**. A semiconductor substrate, an SOI substrate, a glass substrate, or the like is used as the substrate **1190**. The ROM **1199** and the ROM interface **1189** may be provided on a separate chip. Obviously, the CPU illustrated in FIG. 5A is just an example in which the configuration is simplified, and actual CPUs may have various configurations depending on the application.

An instruction input to the CPU through the bus interface **1198** is input to the instruction decoder **1193**, decoded therein, and then input to the ALU controller **1192**, the interrupt controller **1194**, the register controller **1197**, and the timing controller **1195**.

The ALU controller **1192**, the interrupt controller **1194**, the register controller **1197**, and the timing controller **1195** conduct various controls based on the decoded instruction. Specifically, the ALU controller **1192** generates signals for controlling the operation of the ALU **1191**. While the CPU is executing a program, the interrupt controller **1194** processes an interrupt request from an external input/output device or a peripheral circuit on the basis of its priority or a mask state. The register controller **1197** generates an address of the register **1196**, and reads and writes data from and to the register **1196** depending on the state of the CPU.

The timing controller **1195** generates signals for controlling timing of operation of the ALU **1191**, the ALU controller **1192**, the instruction decoder **1193**, the interrupt controller **1194**, and the register controller **1197**. For example, the timing controller **1195** is provided with an internal clock generator for generating an internal clock signal CLK2 based on a reference clock signal CLK1, and supplies the internal clock signal CLK2 to the above-mentioned various circuits.

In the CPU illustrated in FIG. 5A, a memory cell is provided in the register **1196**. The memory cell described above in Embodiment 2 or 3 can be used in the register **1196**.

In the CPU illustrated in FIG. 5A, the register controller **1197** selects operation of holding data in the register **1196** in accordance with an instruction from the ALU **1191**. That is, the register controller **1197** selects whether data is held by a logic element which inverts a logic (logic level) or by a capacitor in the memory cell included in the register **1196**. When data holding by the logic element which inverts a logic (logic level) is selected, a power supply voltage is

supplied to the memory cell in the register 1196. When data holding by the capacitor is selected, the data in the capacitor is rewritten, and supply of the power supply voltage to the memory cell in the register 1196 can be stopped.

The power supply can be stopped by a switching element 5 provided between a memory cell group and a node to which a power supply potential VDD or a power supply potential VSS is supplied, as illustrated in FIG. 5B or FIG. 5C. Circuits illustrated in FIGS. 5B and 5C are described below.

FIGS. 5B and 5C each illustrate an example of a configuration of the storage circuit including the transistor disclosed above in Embodiment 1 as a switching element for controlling supply of a power supply potential to a memory cell.

The storage device illustrated in FIG. 5B includes a switching element 1141 and a memory cell group 1143 including a plurality of memory cells 1142. Specifically, as each of the memory cells 1142, the memory cell described in Embodiment 2 or 3 can be used. Each of the memory cells 1142 included in the memory cell group 1143 is supplied with a high-level power supply potential VDD through the switching element 1141. Further, each of the memory cells 1142 included in the memory cell group 1143 is supplied with a potential of a signal IN and a low-level power supply potential VSS.

As the switching element 1141 in FIG. 5B, the transistor described above in Embodiment 1 is used. The switching of the transistor is controlled by a signal SigA supplied to the gate electrode layer thereof.

Note that FIG. 5B illustrates a configuration in which the switching element 1141 includes only one transistor; however, one embodiment of the present invention is not limited thereto. The switching element 1141 may include a plurality of transistors. In the case where the switching element 1141 includes a plurality of transistors functioning as switching elements, the plurality of transistors may be connected to each other in parallel, in series, or in combination of parallel connection and series connection.

Although the switching element 1141 controls the supply of the high-level power supply potential VDD to each of the memory cells 1142 included in the memory cell group 1143 in FIG. 5B, the switching element 1141 may control the supply of the low-level power supply potential VSS.

FIG. 5C illustrates an example of a storage device in which each of the memory cells 1142 included in the memory cell group 1143 is supplied with the low-level power supply potential VSS through the switching element 1141. The supply of the low-level power supply potential VSS to each of the memory cells 1142 included in the memory cell group 1143 can be controlled by the switching element 1141.

When a switching element is provided between a memory cell group and a node to which the power supply potential VDD or the power supply potential VSS is supplied, data can be held even in the case where operation of a CPU is temporarily stopped and the supply of the power supply voltage is stopped; accordingly, power consumption can be reduced. Specifically, for example, while a user of a personal computer does not input data to an input device such as a keyboard, the operation of the CPU can be stopped, so that power consumption can be reduced.

Although the CPU is given as an example here, the transistor can also be applied to an LSI such as a digital signal processor (DSP), a custom LSI, or a field programmable gate array (FPGA).

The structures, methods, and the like which are described in this embodiment can be combined as appropriate with any of the structures, methods, and the like which are described in the other embodiments.

Further, Table 2 shows a comparison between a spin-MRAM (spin torque transfer MRAM) which is known as a spintronics device and a memory including an oxide semiconductor.

TABLE 2

	Spintronics (MTJ element)	OS/Si
1) Heat resistance	Unstable	Extremely stable (up to 150° C.)
2) Driving method	Current drive	Voltage drive
3) Principle of writing operation	Change Spin Direction of Magnetic Substance	On/off of FET
4) Si LSI	Suitable for bipolar LSI (MOS transistor is preferred in high integration circuit (Bipolar transistor is unsuitable for High Integration); W is large)	Suitable for MOS LSI
5) Overhead	Large (Due to large Joule heat)	Smaller by 2 or 3 or more orders of magnitude (Charge and discharge of parasitic capacitance)
6) Non-volatility	Spin is utilized	Small off-state current is utilized
7) Number of times of reading operation	Unlimited	Unlimited
8) 3D conversion	Difficult (2 layers at most)	Easy (No limitation on the number of layers)
9) Degree of integration (F ²)	4F ² to 15F ²	Depending on the number of layers in 3D conversion
10) Material	Rare-earth element with magnetic property	Oxide semiconductor material
11) Resistance to magnetic field	Low	High

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As shown in Table 2, the memory in which a transistor including an oxide semiconductor and a transistor including silicon are combined is significantly different from the spintronics device in the driving method, the principle of writing operation, the material, and the like.

Further, as shown in Table 2, the memory in which the transistor including an oxide semiconductor and the transistor including silicon are combined has advantages over the spintronics device in many aspects such as the heat resistance, the 3D conversion (stacked-layer structure with three or more layers), and the resistance to a magnetic field. Note that the “overhead” in Table 2 means what is called a power consumed by overhead, which is, for example, power for writing data into a memory portion or the like in a processor.

As described above, the use of the memory including an oxide semiconductor, which has more advantages than the spintronics device, makes it possible to reduce power consumption of a CPU.

The structures, methods, and the like which are described in this embodiment can be combined as appropriate with any of the structures, methods, and the like which are described in the other embodiments.

Embodiment 6

A semiconductor device disclosed in this specification can be applied to a variety of electronic devices (including an amusement machine). Examples of electronic devices include the following: display devices such as televisions and monitors, lighting devices, desktop or laptop personal computers, word processors, image reproduction devices which reproduce still images or moving images stored in recording media such as digital versatile discs (DVDs), portable CD players, radio receivers, tape recorders, headphone stereos, stereos, cordless phone handsets, transceivers, portable wireless devices, cellular phones, car phones, portable game machines, calculators, portable information terminals, electronic notebooks, e-book readers, electronic translators, audio input devices, video cameras, digital still cameras, electric shavers, high-frequency heating appliances such as microwave ovens, electric rice cookers, electric washing machines, electric vacuum cleaners, air-conditioning systems such as air conditioners, dish washing machines, dish drying machines, clothes dryers, futon dryers, electric refrigerators, electric freezers, electric refrigerator-freezers, freezers for preserving DNA, smoke detectors, radiation counters, medical equipment such as dialyzers. Further examples include industrial equipment such as guide lights, traffic lights, belt conveyors, elevators, escalators, industrial robots, and power storage systems. In addition, oil engines, moving objects driven by electric motors using power from non-aqueous secondary batteries, and the like are also included in the range of electronic devices. Examples of the moving objects include electric vehicles (EV), hybrid electric vehicles (HEV) which include both an internal-combustion engine and a motor, plug-in hybrid electric vehicles (PHEV), tracked vehicles in which caterpillar tracks are substituted for wheels of these vehicles, motorized bicycles including motor-assisted bicycles, motorcycles, electric wheelchairs, golf carts, boats or ships, submarines, helicopters, aircrafts, rockets, artificial satellites, space probes, planetary probes, spacecrafts, and the like. Specific examples of these electronic devices are illustrated in FIGS. 6A to 6C.

FIG. 6A illustrates a table 9000 having a display portion. In the table 9000, a display portion 9003 is incorporated in a housing 9001 and an image can be displayed on the display

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portion 9003. Note that the housing 9001 is supported by four leg portions 9002. Further, the housing 9001 is provided with a power cord 9005 for supplying power.

The transistor described in Embodiment 1 can be used in the display portion 9003 so that the electronic device can have high reliability.

The display portion 9003 has a touch-input function. When a user touches displayed buttons 9004 which are displayed on the display portion 9003 of the table 9000 with his/her finger or the like, the user can carry out operation of the screen and input of information. Further, when the table is capable of communicating with other home appliances or control the home appliances, the table 9000 may function as a control device which controls the home appliances by operation on the screen. For example, with the use of a semiconductor device having an image sensing function, the display portion 9003 can have a touch-input function.

Further, the screen of the display portion 9003 can be placed perpendicular to a floor with a hinge provided for the housing 9001; thus, the table 9000 can also be used as a television device. When a television device having a large screen is set in a small room, an open space is reduced; however, when a display portion is incorporated in a table, a space in the room can be efficiently used.

FIG. 6B illustrates a portable music player, which includes, in a main body 3021, a display portion 3023, a fixing portion 3022 with which the main body is worn on the ear, an operation button 3024, an external connection port 3025, and the like. The portable music player may include a speaker. When the transistor described in Embodiment 1 or any of the memories or logic circuits described in Embodiments 2 to 4 is applied to a memory, a CPU, or the like included in the main body 3021, a portable music player (PDA) whose power consumption is reduced can be provided.

Furthermore, when the portable music player illustrated in FIG. 6B has an antenna, a microphone function, or a wireless communication function and is used with a mobile phone, a user can talk on the phone wirelessly in a hands-free way while driving a car or the like.

FIG. 6C illustrates a computer, which includes a main body 9201 including a CPU, a housing 9202, a display portion 9203, a keyboard 9204, an external connection port 9205, a pointing device 9206, and the like. The computer is manufactured using a semiconductor device manufactured according to one embodiment of the present invention for the display portion 9203. When the CPU described in Embodiment 5 is used, power consumption of the computer can be reduced.

FIGS. 7A and 7B illustrate a foldable tablet terminal. The tablet terminal is opened in FIG. 7A. The tablet terminal includes a housing 9630, a display portion 9631a, a display portion 9631b, a display mode switch 9034, a power switch 9035, a power saver switch 9036, a clasp 9033, and an operation switch 9038.

In the portable device illustrated in FIGS. 7A and 7B, a memory such as an SRAM or a DRAM is used for temporarily storing image data or the like. For example, the semiconductor device described in Embodiment 2 or 3 can be used as a memory. By employing the semiconductor device described in the above embodiment for the memory, data can be written and read at high speed and held for a long time, and power consumption can be sufficiently reduced.

Part of the display portion 9631a can be a touch panel region 9632a and data can be input when a displayed operation key 9638 is touched. Although a structure in which a half region in the display portion 9631a has only a display

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function and the other half region has a touch panel function is shown as an example, the display portion **9631a** is not limited to this structure. The whole region in the display portion **9631a** may have a touch panel function. For example, the display portion **9631a** can display keyboard buttons in its whole region to serve as a touch panel, and the display portion **9631b** can be used as a display screen.

As in the display portion **9631a**, part of the display portion **9631b** can be a touch panel region **9632b**. When a keyboard display switching button **9639** displayed on the touch panel is touched with a finger, a stylus, or the like, a keyboard can be displayed on the display portion **9631b**.

Touch input can be performed in the touch panel region **9632a** and the touch panel region **9632b** at the same time.

The display mode switch **9034** can switch the display between portrait mode, landscape mode, and the like, and between monochrome display and color display, for example. The power saver switch **9036** can control display luminance in accordance with the amount of external light in use of the tablet terminal detected by an optical sensor incorporated in the tablet terminal. In addition to the optical sensor, another detection device including a sensor for detecting inclination, such as a gyroscope or an acceleration sensor, may be incorporated in the tablet terminal.

Note that FIG. 7A shows an example in which the display portion **9631a** and the display portion **9631b** have the same display area; however, without limitation, one of the display portions may be different from the other display portion in size and display quality. For example, one display panel may be capable of higher-definition display than the other display panel.

The tablet terminal is closed in FIG. 7B. The tablet terminal includes the housing **9630**, a solar cell **9633**, a charge and discharge control circuit **9634**, a battery **9635**, and a DCDC converter **9636**. In FIG. 7B, a structure including the battery **9635** and the DCDC converter **9636** is illustrated as an example of the charge and discharge control circuit **9634**.

Since the tablet terminal is foldable, the housing **9630** can be closed when the tablet terminal is not used. As a result, the display portion **9631a** and the display portion **9631b** can be protected; thus, a tablet terminal which has excellent durability and excellent reliability in terms of long-term use can be provided.

In addition, the tablet terminal illustrated in FIGS. 7A and 7B can have a function of displaying a variety of kinds of data (e.g., a still image, a moving image, and a text image), a function of displaying a calendar, a date, the time, or the like on the display portion, a touch-input function of operating or editing the data displayed on the display portion by touch input, a function of controlling processing by a variety of kinds of software (programs), and the like.

The solar cell **9633** provided on a surface of the tablet terminal can supply power to the touch panel, the display portion, a video signal processing portion, or the like. Note that the solar cell **9633** can be provided on one or both surfaces of the housing **9630**, and the battery **9635** can be charged efficiently. The use of a lithium ion battery as the battery **9635** is advantageous in downsizing or the like.

The structure and the operation of the charge and discharge control circuit **9634** illustrated in FIG. 7B will be described with reference to a block diagram in FIG. 7C. The solar cell **9633**, the battery **9635**, the DCDC converter **9636**, a converter **9637**, switches SW1 to SW3, and a display portion **9631** are illustrated in FIG. 7C, and the battery **9635**, the DCDC converter **9636**, the converter **9637**, and the

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switches SW1 to SW3 correspond to the charge and discharge control circuit **9634** illustrated in FIG. 7B.

First, an example of the operation in the case where power is generated by the solar cell **9633** using external light is described. The voltage of power generated by the solar cell is stepped up or down by the DCDC converter **9636** so that the power has a voltage for charging the battery **9635**. Then, when the power from the solar cell **9633** is used for the operation of the display portion **9631**, the switch SW1 is turned on and the voltage of the power is stepped up or down by the converter **9637** so as to be a voltage needed for the display portion **9631**. In addition, when display on the display portion **9631** is not performed, the switch SW1 is turned off and the switch SW2 is turned on so that the battery **9635** may be charged.

Note that the solar cell **9633** is described as an example of a power generation means; however, without limitation, the battery **9635** may be charged using another power generation means such as a piezoelectric element or a thermoelectric conversion element (Peltier element). For example, a non-contact electric power transmission module which transmits and receives power wirelessly (without contact) to charge the battery **9635**, or a combination of the solar cell **9633** and another means for charge may be used.

In a television device **8000** in FIG. 8A, a display portion **8002** is incorporated in a housing **8001**. The display portion **8002** displays an image and a speaker portion **8003** can output sound. The transistor described in Embodiment 1 can be used in the display portion **8002**.

A semiconductor display device such as a liquid crystal display device, a light-emitting device in which a light-emitting element such as an organic EL element is provided in each pixel, an electrophoresis display device, a digital micromirror device (DMD), a plasma display panel (PDP), or the like can be used in the display portion **8002**.

The television device **8000** may be provided with a receiver, a modem, and the like. With the receiver, the television device **8000** can receive general television broadcasting. Furthermore, when the television device **8000** is connected to a communication network by wired or wireless connection via the modem, one-way (from a transmitter to a receiver) or two-way (between a transmitter and a receiver, between receivers, or the like) data communication can be performed.

In addition, the television device **8000** may include a CPU for performing information communication or a memory. Any of the memories, logic circuits, or CPU described in Embodiments 2 to 5 can be used in the television device **8000**.

In FIG. 8A, an air conditioner including an indoor unit **8200** and an outdoor unit **8204** is an example of an electronic device including the CPU of Embodiment 5. Specifically, the indoor unit **8200** includes a housing **8201**, a ventilation duct **8202**, a CPU **8203**, and the like. FIG. 8A shows the case where the CPU **8203** is provided in the indoor unit **8200**; the CPU **8203** may be provided in the outdoor unit **8204**. Alternatively, the CPU **8203** may be provided in both the indoor unit **8200** and the outdoor unit **8204**. Since the CPU described in Embodiment 5 is formed using an oxide semiconductor, an air conditioner which has excellent heat resistance and high reliability can be provided with the use of the CPU.

In FIG. 8A, an electric refrigerator-freezer **8300** is an example of an electronic device which is provided with the CPU formed using an oxide semiconductor. Specifically, the electric refrigerator-freezer **8300** includes a housing **8301**, a refrigerator door **8302**, a freezer door **8303**, a CPU **8304**,

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and the like. The CPU **8304** is provided in the housing **8301** in FIG. **8A**. When the CPU described in Embodiment 5 is used as the CPU **8304** of the electric refrigerator-freezer **8300**, power saving can be achieved.

FIGS. **8B** and **8C** illustrates an example of an electric vehicle which is an example of an electronic device. An electric vehicle **9700** is equipped with a secondary battery **9701**. The output of power of the secondary battery **9701** is controlled by a control circuit **9702** and the power is supplied to a driving device **9703**. The control circuit **9702** is controlled by a processing unit **9704** including a ROM, a RAM, a CPU, or the like which is not illustrated. When the CPU described in Embodiment 5 is used as the CPU in the electric vehicle **9700**, power saving can be achieved.

The driving device **9703** includes a DC motor or an AC motor either alone or in combination with an internal-combustion engine. The processing unit **9704** outputs a control signal to the control circuit **9702** based on input data such as data of operation (e.g., acceleration, deceleration, or stop) by a driver or data during driving (e.g., data on an upgrade or a downgrade, or data on a load on a driving wheel) of the electric vehicle **9700**. The control circuit **9702** adjusts the electric energy supplied from the secondary battery **9701** in accordance with the control signal of the processing unit **9704** to control the output of the driving device **9703**. In the case where the AC motor is mounted, although not illustrated, an inverter which converts direct current into alternate current is also incorporated.

The structures, methods, and the like which are described in this embodiment can be combined as appropriate with any of the structures, methods, and the like which are described in the other embodiments.

EXPLANATION OF REFERENCE

100: substrate, **102**: gate electrode layer, **103**: insulating layer, **104**: insulating layer, **106**: oxide semiconductor stacked layers, **106a**: oxide semiconductor layer, **106b**: oxide semiconductor layer, **108**: conductive film, **108a**: source electrode layer, **108b**: drain electrode layer, **110**: insulating layer, **112**: gate electrode layer, **120**: transistor, **160**: transistor, **162**: transistor, **164**: capacitor, **200**: substrate, **202a**: electrode layer, **202b**: gate electrode layer, **203**: insulating layer, **204**: insulating layer, **206**: element isolation insulating layer, **208**: gate insulating layer, **210**: gate electrode layer, **212a**: electrode layer, **212b**: electrode layer, **214**: impurity region, **216**: channel formation region, **218**: sidewall insulating layer, **220**: high-concentration impurity region, **222a**: electrode layer, **222b**: wiring layer, **224**: intermetallic compound region, **228**: insulating layer, **230**: insulating layer, **232**: insulating layer, **234**: insulating layer, **236**: insulating layer, **244**: oxide semiconductor stacked layers, **244a**: oxide semiconductor layer, **244b**: oxide semiconductor layer, **250**: memory cell, **251**: memory cell array, **251a**: memory cell array, **251b**: memory cell array, **253**: conductive layer, **254**: capacitor, **256**: wiring layer, **258**: peripheral circuit, **260**: gate insulating layer, **262**: gate electrode layer, **268a**: electrode layer, **268b**: electrode layer, **302**: insulating layer, **306**: oxide semiconductor layer, **308a**: source electrode layer, **308b**: drain electrode layer, **310**: gate insulating layer, **320**: transistor, **402**: oxygen-excess region, **405a**: source electrode layer, **405b**: drain electrode layer, **400**: oxygen, **800**: substrate, **801**: transistor, **802**: transistor, **803**: transistor, **804**: transistor, **806**: element isolation insulating layer, **811**: transistor, **812**: transistor, **813**:

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transistor, **814**: transistor, **821**: gate electrode layer, **825**: electrode layer, **826**: insulating layer, **830**: insulating layer, **831**: electrode layer, **832**: wiring layer, **833**: insulating layer, **834**: wiring layer, **835**: electrode layer, **836**: insulating layer, **839**: insulating layer, **840**: gate electrode layer, **841a**: gate electrode layer, **841b**: electrode layer, **843**: gate insulating layer, **845**: electrode layer, **1141**: switching element, **1142**: memory cell, **1143**: memory cell group, **1189**: ROM interface, **1190**: substrate, **1191**: ALU, **1192**: ALU controller, **1193**: instruction decoder, **1194**: interrupt controller, **1195**: timing controller, **1196**: register, **1197**: register controller, **1198**: bus interface, **1199**: ROM, **3021**: main body, **3022**: fixing portion, **3023**: display portion, **3024**: operation button, **3025**: external connection port, **8000**: television device, **8001**: housing, **8002**: display portion, **8003**: speaker portion, **8200**: indoor unit, **8201**: housing, **8202**: ventilation duct, **8203**: CPU, **8204**: outdoor unit, **8300**: electric refrigerator-freezer, **8301**: housing, **8302**: refrigerator door, **8303**: freezer door, **8304**: CPU, **9000**: table, **9001**: housing, **9002**: leg portions, **9003**: display portion, **9004**: displayed buttons, **9005**: power cord, **9033**: clasp, **9034**: switch, **9035**: power switch, **9036**: switch, **9038**: operation switch, **9201**: main body, **9202**: housing, **9203**: display portion, **9204**: keyboard, **9205**: external connection port, **9206**: pointing device, **9630**: housing, **9631**: display portion, **9631a**: display portion, **9631b**: display portion, **9632a**: region, **9632b**: region, **9633**: solar cell, **9634**: charge and discharge control circuit, **9635**: battery, **9636**: DCDC converter, **9637**: converter, **9638**: operation key, **9639**: button, **9700**: electric vehicle, **9701**: secondary battery, **9702**: control circuit, **9703**: driving device, **9704**: processing unit.

This application is based on Japanese Patent Application serial no. 2012-091539 filed with Japan Patent Office on Apr. 13, 2012, the entire contents of which are hereby incorporated by reference.

The invention claimed is:

1. A semiconductor device comprising:

a first transistor;
a second transistor; and
a capacitor,

wherein the first transistor comprises a first channel formation region including silicon,

wherein the second transistor comprises a second channel formation region,

wherein a gate electrode of the first transistor is electrically connected to one of a source electrode and a drain electrode of the second transistor and one electrode of the capacitor,

wherein a first insulating layer is provided over the first channel formation region,

wherein a first conductive layer in contact with a top surface of the first insulating layer comprises a region configured to function as the gate electrode of the first transistor,

wherein a second insulating layer is provided over the first conductive layer,

wherein a second conductive layer in contact with a top surface of the second insulating layer is configured to function as a first gate electrode of the second transistor,

wherein a third insulating layer is provided in contact with a top surface of the second conductive layer,

wherein a fourth insulating layer is provided in contact with a top surface of the third insulating layer,

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wherein an oxide semiconductor layer in contact with a top surface of the fourth insulating layer comprises the second channel formation region,
 wherein a fifth insulating layer is provided over the oxide semiconductor layer,
 wherein a third conductive layer in contact with a top surface of the fifth insulating layer is configured to function as a second gate electrode of the second transistor,
 wherein a fourth conductive layer in contact with a top surface of the oxide semiconductor layer comprises a region configured to function as the one of the source electrode and the drain electrode of the second transistor,
 wherein a fifth conductive layer in contact with the top surface of the oxide semiconductor layer comprises a region configured to function as the other of the source electrode and the drain electrode of the second transistor,
 wherein a sixth conductive layer in contact with the top surface of the fifth insulating layer overlaps with the fourth conductive layer,
 wherein a seventh conductive layer provided over the sixth conductive layer is electrically connected to the fifth conductive layer,
 wherein an eighth conductive layer in contact with the top surface of the second insulating layer overlaps with the fourth conductive layer,
 wherein the seventh conductive layer comprises a region overlapping the second channel formation region and a region overlapping the capacitor,
 wherein the second conductive layer and the eighth conductive layer comprise a same material, and
 wherein the third conductive layer and the sixth conductive layer comprise a same material.

2. A semiconductor device comprising:
 a first transistor; and
 a second transistor,
 wherein the first transistor comprises a first channel formation region including silicon,
 wherein the second transistor comprises a second channel formation region,
 wherein a gate electrode of the first transistor is electrically connected to one of a source electrode and a drain electrode of the second transistor,
 wherein a first insulating layer is provided over the first channel formation region,
 wherein a first conductive layer in contact with a top surface of the first insulating layer comprises a region configured to function as the gate electrode of the first transistor,
 wherein a second insulating layer is provided over the first conductive layer,
 wherein a second conductive layer in contact with a top surface of the second insulating layer is configured to function as a first gate electrode of the second transistor,
 wherein a third insulating layer is provided over the second conductive layer,
 wherein an oxide semiconductor layer in contact with a top surface of the third insulating layer comprises the second channel formation region,
 wherein a fourth insulating layer is provided over the oxide semiconductor layer,

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wherein a third conductive layer in contact with a top surface of the fourth insulating layer is configured to function as a second gate electrode of the second transistor,
 wherein a fourth conductive layer in contact with a top surface of the oxide semiconductor layer comprises a region configured to function as the one of the source electrode and the drain electrode of the second transistor,
 wherein a fifth conductive layer in contact with the top surface of the oxide semiconductor layer comprises a region configured to function as the other of the source electrode and the drain electrode of the second transistor,
 wherein a sixth conductive layer in contact with the top surface of the fourth insulating layer overlaps with the fourth conductive layer,
 wherein a seventh conductive layer in contact with the top surface of the second insulating layer overlaps with the fourth conductive layer,
 wherein the second conductive layer and the seventh conductive layer comprise a same material, and
 wherein the third conductive layer and the sixth conductive layer comprise a same material.

3. A semiconductor device comprising:
 a first transistor;
 a second transistor; and
 a capacitor,
 wherein the first transistor comprises a first channel formation region including silicon,
 wherein the second transistor comprises a second channel formation region,
 wherein a gate electrode of the first transistor is electrically connected to one of a source electrode and a drain electrode of the second transistor and one electrode of the capacitor,
 wherein a first insulating layer is provided over the first channel formation region,
 wherein a first conductive layer in contact with a top surface of the first insulating layer comprises a region configured to function as the gate electrode of the first transistor,
 wherein a second insulating layer is provided over the first conductive layer,
 wherein a second conductive layer in contact with a top surface of the second insulating layer is configured to function as a first gate electrode of the second transistor,
 wherein a third insulating layer is provided in contact with a top surface of the second conductive layer,
 wherein a fourth insulating layer is provided in contact with a top surface of the third insulating layer,
 wherein an oxide semiconductor layer in contact with a top surface of the fourth insulating layer comprises the second channel formation region,
 wherein a fifth insulating layer is provided over the oxide semiconductor layer,
 wherein a third conductive layer in contact with a top surface of the fifth insulating layer is configured to function as a second gate electrode of the second transistor,
 wherein a fourth conductive layer in contact with a top surface of the oxide semiconductor layer comprises a region configured to function as the one of the source electrode and the drain electrode of the second transistor,

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wherein a fifth conductive layer in contact with the top surface of the oxide semiconductor layer comprises a region configured to function as the other of the source electrode and the drain electrode of the second transistor,
 wherein a sixth conductive layer in contact with the top surface of the fifth insulating layer overlaps with the fourth conductive layer,
 wherein a seventh conductive layer provided over the sixth conductive layer is electrically connected to the fifth conductive layer,
 wherein an eighth conductive layer in contact with the top surface of the second insulating layer overlaps with the fourth conductive layer,
 wherein the seventh conductive layer comprises a region overlapping the second channel formation region and a region overlapping the capacitor,
 wherein the second conductive layer and the eighth conductive layer comprise a same material,
 wherein the third conductive layer and the sixth conductive layer comprise a same material, and
 wherein a thickness of the fourth insulating layer is greater than a thickness of the third insulating layer.

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4. The semiconductor device according to claim 2, wherein a thickness of the fourth insulating layer is greater than a thickness of the third insulating layer.

5. The semiconductor device according to claim 1, wherein the seventh conductive layer comprises a region overlapping with the first channel formation region.

6. The semiconductor device according to claim 2, wherein the seventh conductive layer comprises a region overlapping with the first channel formation region.

7. The semiconductor device according to claim 3, wherein the seventh conductive layer comprises a region overlapping with the first channel formation region.

8. The semiconductor device according to claim 1, wherein the oxide semiconductor layer comprises In, Ga, and Zn.

9. The semiconductor device according to claim 2, wherein the oxide semiconductor layer comprises In, Ga, and Zn.

10. The semiconductor device according to claim 3, wherein the oxide semiconductor layer comprises In, Ga, and Zn.

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